

第01页 | 行0001-0050 | web/src/App.vue ... web/src/api/admin.mjs

```
0001 <template>
0002   <NavBar />
0003   <router-view></router-view>
0004 </template>
0005
0006 <script>
0007   import NavBar from './components/NavBar.vue';
0008   import "bootstrap/dist/css/bootstrap.min.css";
0009   import "bootstrap/dist/js/bootstrap";
0010   export default{
0011     components: {
0012       NavBar
0013     },
0014   }
0015 </script>
0016
0017 <style>
0018   body{
0019     background-image: url("@/assets/images/brck.avif");
0020     background-size: cover;
0021   }
0022 </style>
0023   import $ from 'jquery'
0024
0025   const API_BASE = 'http://127.0.0.1:3000'
0026
0027   const request = options => new Promise((resolve, reject) => {
0028     const { token, ...ajaxOptions } = options
0029
0030     $.ajax({
0031       ...ajaxOptions,
0032       url: `${API_BASE}${ajaxOptions.url}`,
0033       headers: {
0034         ...ajaxOptions.headers,
0035         Authorization: `Bearer ${token}`
0036       },
0037       success: resolve,
0038       error: reject
0039     })
0040   })
0041
0042   export const requestErrorMessage = error => {
0043     if (error?.responseJSON?.error_message) {
0044       return error.responseJSON.error_message
0045     }
0046     if (error?.status === 401) {
0047       return '登录状态已失效, 请重新登录'
0048     }
0049     if (error?.status === 0) {
0050       return '无法连接到服务器'
```

第 02 页 | 行 0051-0100 | web/src/api/admin.mjs

```
0051   }
0052   return error?.statusText || '请求失败, 请稍后重试'
0053 }
0054
0055 export const adminApi = {
0056   listUsers: token => request({
0057     url: '/manage/user/gitlist/',
0058     type: 'GET',
0059     token
0060   }),
0061
0062   updateBanned: (token, username, status) => request({
0063     url: '/user/account/banned/',
0064     type: 'GET',
0065     token,
0066     data: { username, status }
0067   }),
0068
0069   listLoginRecords: token => request({
0070     url: '/manage/record/login/',
0071     type: 'GET',
0072     token
0073   }),
0074
0075   listTopics: token => request({
0076     url: '/oj/topic/getlist/',
0077     type: 'GET',
0078     token
0079   }),
0080
0081   getTopic: (token, id) => request({
0082     url: '/oj/topic/get/',
0083     type: 'GET',
0084     token,
0085     data: { id }
0086   }),
0087
0088   addTopic: (token, payload) => request({
0089     url: '/oj/topic/add/',
0090     type: 'POST',
0091     token,
0092     data: payload
0093   }),
0094
0095   updateTopic: (token, payload) => request({
0096     url: '/oj/topic/updata/',
0097     type: 'POST',
0098     token,
0099     data: payload
0100   }),
```

第03页 | 行0101-0150 | web/src/api/admin.mjs ... web/src/assets/scripts/AcGameObject.js

```
0101
0102   removeTopic: (token, topicId) => request({
0103     url: '/oj/topic/remove/',
0104     type: 'POST',
0105     token,
0106     data: { topic_id: topicId }
0107   }),
0108
0109   getEvaluates: (token, topicId) => request({
0110     url: '/manage/evaluate/get/',
0111     type: 'POST',
0112     token,
0113     data: { topic_id: topicId }
0114   }),
0115
0116   updateEvaluates: (token, payload) => request({
0117     url: '/manage/evaluate/updata/',
0118     type: 'POST',
0119     token,
0120     contentType: 'application/json; charset=UTF-8',
0121     data: JSON.stringify(payload)
0122   })
0123 }
0124 export function createBlogImageFormData(blogId, file, FormDataClass = FormData) {
0125   const data = new FormDataClass()
0126   data.append('blog_id', String(blogId)); data.append('file', file)
0127   return data
0128 }
0129
0130 export async function uploadBlogImage(blogId, file, token) {
0131   const response = await fetch('http://127.0.0.1:3000/user/blog/image/upload/', {
0132     method: 'POST', headers: { Authorization: `Bearer ${token}` }, body: createBlogImageFormData(blogId, file)
0133   })
0134   const data = await response.json()
0135   if (!response.ok || data.error_message !== 'success') throw new Error(data.error_message || '图片上传失败')
0136   return data.url
0137 }
0138 const AC_GAME_OBJECTS = [];
0139
0140 export class AcGameObject{
0141   constructor() {
0142     AC_GAME_OBJECTS.push(this);
0143     this.timedelta = 0;
0144     this.has_called_start = false;
0145   }
0146
0147   start(){
0148
0149   }
0150 }
```

第04页 | 行0151-0200 | web/src/assets/scripts/AcGameObject.js ... web/src/assets/scripts/GameMap.js

```

0151     update(){
0152     }
0153     }
0154     on_destroy(){
0155     }
0156     }
0157     destroy() {
0158         this.on_destroy();
0159
0160         for(let i in AC_GAME_OBJECTS){
0161             const obj = AC_GAME_OBJECTS[i];
0162             if(obj == this){
0163                 AC_GAME_OBJECTS.splice(i);
0164                 break;
0165             }
0166         }
0167     }
0168 }
0169 let last_timestamp;
0170 const step = timestamp => {
0171     for(let obj of AC_GAME_OBJECTS){
0172         if(!obj.has_called_start){
0173             obj.has_called_start = true;
0174             obj.start();
0175         }
0176         else{
0177             obj.timedelta = timestamp - last_timestamp;
0178             obj.update();
0179         }
0180     }
0181     last_timestamp = timestamp;
0182     requestAnimationFrame(step)
0183 }
0184
0185
0186
0187 requestAnimationFrame(step)
0188 export class Cell{
0189     constructor(r, c){
0190         this.r = r;
0191         this.c = c;
0192         this.x = c + 0.5;
0193         this.y = r + 0.5
0194     }
0195 }
0196 import { AcGameObject } from './AcGameObject';
0197 import { Wall } from './Wall';
0198 import { Snake } from './Snake';
0199 export class GameMap extends AcGameObject{
0200     constructor(ctx,parent){

```

第05页 | 行 0201-0250 | web/src/assets/scripts/GameMap.js

```

0201         super();
0202
0203         this.ctx = ctx;
0204         this.parent = parent;
0205         this.L = 0;
0206
0207         this.rows = 13;
0208         this.cols = 14;
0209
0210         this.inner_walls_count = 20;
0211         this.walls = [];
0212
0213         this.snakes = [
0214             new Snake({id: 0, color: "#4876EC", r: this.rows - 2, c: 1}, this),
0215             new Snake({id: 1, color: "#F94848", r: 1, c: this.cols - 2}, this),
0216         ];
0217
0218     }
0219
0220     check_connectivity(g, sx, sy, tx, ty){ // 判断左下角和右上角是否连通
0221         if(sx == tx && sy == ty) return true;
0222         g[sx][sy] = true;
0223
0224         let dx = [-1, 0, 1, 0], dy = [0, 1, 0, -1];
0225         for(let i = 0; i < 4; i++){
0226             let x = sx + dx[i], y = sy + dy[i];
0227             if(!g[x][y] && this.check_connectivity(g, x, y, tx, ty)){
0228                 return true;
0229             }
0230         }
0231         return false;
0232     }
0233
0234     create_walls(){ // 创建障碍物
0235         const g = [];
0236         for(let r = 0; r < this.rows; r++){
0237             g[r] = [];
0238             for(let c = 0; c < this.cols; c++){
0239                 g[r][c] = false;
0240             }
0241         }
0242
0243         for(let r = 0; r < this.rows; r++){
0244             g[r][0] = g[r][this.cols - 1] = true;
0245         }
0246
0247         for(let c = 0; c < this.cols; c++){
0248             g[0][c] = g[this.rows - 1][c] = true;
0249         }
0250

```

第06页 | 行0251-0300 | web/src/assets/scripts/GameMap.js

```

0251     for(let i = 0; i < this.inner_walls_count / 2; i++){
0252         for(let j = 0; j < 1000; j++){
0253             let r = parseInt(Math.random() * this.rows);
0254             let c = parseInt(Math.random() * this.cols);
0255             if(r == this.rows - 2 && c == 1 || r == 1 && c == this.cols - 2){
0256                 continue;
0257             }
0258             if(g[r][c] || g[c][r]){
0259                 continue;
0260             }
0261             else{
0262                 g[r][c] = g[this.rows - r - 1][this.cols - c - 1] = true;
0263                 break;
0264             }
0265         }
0266     }
0267     const copy_g = JSON.parse(JSON.stringify(g));
0268     if(!this.check_connectivity(copy_g, this.rows - 2, 1, 1, this.cols - 2))
0269         return false;
0270
0271     for(let r = 0; r < this.rows; r++){
0272         for(let c = 0; c < this.cols; c++){
0273             if(g[r][c]){
0274                 this.walls.push(new Wall(r, c, this));
0275             }
0276         }
0277     }
0278     return true;
0279 }
0280
0281 add_listening_events(){
0282     this.ctx.canvas.focus();
0283
0284     const [snake0, snake1] = this.snakes;
0285     this.ctx.canvas.addEventListener("keydown", e => { // 从键盘中读入方向, 并且不会回头
0286         if(e.key === 'w') snake0.set_direction(0);
0287         else if(e.key === 'd') snake0.set_direction(1);
0288         else if(e.key === 's') snake0.set_direction(2);
0289         else if(e.key === 'a') snake0.set_direction(3);
0290         else if(e.key === 'ArrowUp') snake1.set_direction(0);
0291         else if(e.key === 'ArrowRight') snake1.set_direction(1);
0292         else if(e.key === 'ArrowDown') snake1.set_direction(2);
0293         else if(e.key === 'ArrowLeft') snake1.set_direction(3);
0294     });
0295 }
0296
0297 start(){
0298     for(let i = 0; i < 1000; i++){ // 重复创建1000次障碍物, 直到合法
0299         if(this.create_walls()){
0300             break;

```

第 07 页 | 行 0301-0350 | web/src/assets/scripts/GameMap.js

```

0301         }
0302     }
0303
0304     this.add_listening_events();
0305 }
0306 update_size(){ //算出地图的相对长宽
0307     this.L = Math.min(Math.trunc(this.parent.clientWidth/this.cols), Math.trunc(this.parent.clientHeight/this.rows));
0308     this.ctx.canvas.width = this.L * this.cols;
0309     this.ctx.canvas.height = this.L * this.rows;
0310 }
0311
0312 next_step(){
0313     for(const snake of this.snakes){
0314         snake.next_step();
0315     }
0316 }
0317
0318 check_ready(){
0319     for(const snake of this.snakes){
0320         if(snake.status !== "idle") return false;
0321         if(snake.direction === -1) return false;
0322     }
0323     return true;
0324 }
0325
0326 check_valid(cell){
0327     for(const wall of this.walls){
0328         if(wall.r === cell.r && wall.c === cell.c){
0329             return false;
0330         }
0331     }
0332
0333     for(const snake of this.snakes){
0334         let k = snake.cells.length;
0335         if(!snake.check_tall_increasing()){
0336             k --;
0337         }
0338         for(let i = 0; i < k; i ++){
0339             if(snake.cells[i].r === cell.r && snake.cells[i].c === cell.c){
0340                 return false;
0341             }
0342         }
0343     }
0344     return true;
0345 }
0346
0347 update(){
0348     this.update_size();
0349     if(this.check_ready()){
0350         this.next_step();

```

第 08 页 | 行 0351-0400 | web/src/assets/scripts/GameMap.js ... web/src/assets/scripts/Snake.js

```

0351     }
0352     this.render();
0353 }
0354
0355 render(){ //创建地图
0356     const color_even = "#AAD751", color_odd = "#8eca18ff";
0357     for(let r = 0; r < this.rows; r++){
0358         for(let c = 0; c < this.cols; c++){
0359             if((r + c) % 2 == 0){
0360                 this.ctx.fillStyle = color_even;
0361             }
0362             else{
0363                 this.ctx.fillStyle = color_odd;
0364             }
0365             this.ctx.fillRect(c * this.L, r * this.L, this.L, this.L);
0366         }
0367     }
0368 }
0369 }
0370 import { AcGameObject } from "./AcGameObject";
0371 import { Cell } from "./Cell";
0372
0373 export class Snake extends AcGameObject{
0374     constructor(info, gamemap){
0375         super();
0376
0377         this.id = info.id;
0378         this.color = info.color;
0379         this.gamemap = gamemap;
0380
0381         this.cells = [new Cell(info.r, info.c)];
0382         this.next_cell = null;
0383
0384
0385
0386         this.speed = 5;
0387         this.direction = -1;
0388         this.status = "idle";
0389
0390         this.dr = [-1, 0, 1, 0];
0391         this.dc = [0, 1, 0, -1];
0392
0393         this.step = 0;
0394         this.eps = 1e-2;
0395
0396
0397         this.eye_direction = 0;
0398         if(this.id === 1){
0399             this.eye_direction = 2;
0400         }

```


第09页 | 行0401-0450 | web/src/assets/scripts/Snake.js

```
0401
0402     this.eye_dx = [ // 蛇眼睛的x偏移量
0403         [-1, 1],
0404         [1, 1],
0405         [1, -1],
0406         [-1, -1],
0407     ];
0408     this.eye_dy = [ // 蛇眼睛的y偏移量
0409         [-1, -1],
0410         [-1, 1],
0411         [1, 1],
0412         [1, -1],
0413     ];
0414 }
0415
0416 start(){
0417
0418 }
0419
0420 set_direction(d){
0421     this.direction = d;
0422 }
0423
0424 check_tall_increasing(){
0425     if(this.step <= 10) return true;
0426     if(this.step % 3 === 1) return true;
0427     return false;
0428 }
0429
0430 next_step(){
0431     const d = this.direction;
0432     this.next_cell = new Cell(this.cells[0].r + this.dr[d], this.cells[0].c + this.dc[d]);
0433     this.eye_direction = d;
0434     this.direction = -1;
0435     this.status = "move";
0436     this.step ++;
0437     const k = this.cells.length;
0438     for(let i = k; i > 0; i --){
0439         this.cells[i] = JSON.parse(JSON.stringify(this.cells[i - 1]));
0440     }
0441
0442     if(!this.gamemap.check_valid(this.next_cell)){
0443         this.status = "die";
0444     }
0445 }
0446
0447
0448 update_move(){
0449     const dx = this.next_cell.x - this.cells[0].x;
0450     const dy = this.next_cell.y - this.cells[0].y;
```

第 10 页 | 行 0451-0500 | web/src/assets/scripts/Snake.js

```

0451     const distance = Math.sqrt(dx * dx + dy * dy);
0452
0453     if(distance < this.eps){
0454         this.cells[0] = this.next_cell;
0455         this.next_cell = null;
0456         this.status = "idle";
0457
0458         if(!this.check_tall_increasing()){
0459             this.cells.pop();
0460         }
0461     }else{
0462         const move_distance = this.speed * this.timedelta / 1000;
0463         this.cells[0].x += move_distance * dx / distance;
0464         this.cells[0].y += move_distance * dy / distance;
0465
0466         if(!this.check_tall_increasing()){
0467             const k = this.cells.length;
0468             const tail = this.cells[k - 1], tail_target = this.cells[k - 2];
0469             const tail_dx = tail_target.x - tail.x;
0470             const tail_dy = tail_target.y - tail.y;
0471             tail.x += move_distance * tail_dx / distance;
0472             tail.y += move_distance * tail_dy / distance;
0473         }
0474     }
0475 }
0476
0477 update(){
0478     if(this.status === 'move'){
0479         this.update_move();
0480     }
0481     this.render();
0482 }
0483
0484 render(){
0485     const L = this.gamemap.L;
0486     const ctx = this.gamemap.ctx;
0487
0488     ctx.fillStyle = this.color;
0489     if(this.status === "die"){
0490         ctx.fillStyle = "white";
0491     }
0492
0493     for(const cell of this.cells){
0494         ctx.beginPath();
0495         ctx.arc(cell.x * L, cell.y * L, L * 0.4, 0, Math.PI * 2);
0496         ctx.fill();
0497     }
0498
0499     for(let i = 1; i < this.cells.length; i++){
0500

```

第 11 页 | 行 0501-0550 | web/src/assets/scripts/Snake.js ... web/src/assets/scripts/Wall.js

```

0501     const a = this.cells[i - 1], b = this.cells[i];
0502     if(Math.abs(a.x - b.x) < this.eps && Math.abs(a.y - b.y) < this.eps) continue;
0503     if(Math.abs(a.x - b.x) < this.eps){
0504         ctx.fillRect((a.x - 0.4) * L, Math.min(a.y, b.y) * L, L * 0.8, Math.abs(a.y - b.y) * L );
0505     }
0506     else{
0507         ctx.fillRect(Math.min(a.x, b.x) * L, (a.y - 0.4) * L, Math.abs(a.x - b.x) * L, L * 0.8 );
0508     }
0509 }
0510
0511 ctx.fillStyle = "black";
0512 for(let i = 0; i < 2; i++){
0513     if(this.status === "die"){ //将死去蛇的眼睛变成X
0514         const eye_x = (this.cells[0].x + this.eye_dx[this.eye_direction][i] * 0.15 - 0.05) * L;
0515         const eye_y = (this.cells[0].y + this.eye_dy[this.eye_direction][i] * 0.15 - 0.05) * L;
0516         ctx.beginPath();
0517         ctx.moveTo(eye_x, eye_y);
0518         ctx.lineTo(eye_x + L * 0.1, eye_y + L * 0.1);
0519         ctx.moveTo(eye_x + L * 0.1, eye_y);
0520         ctx.lineTo(eye_x, eye_y + L * 0.1);
0521         ctx.stroke();
0522         continue;
0523     }
0524     const eye_x = (this.cells[0].x + this.eye_dx[this.eye_direction][i] * 0.15) * L;
0525     const eye_y = (this.cells[0].y + this.eye_dy[this.eye_direction][i] * 0.15) * L;
0526     ctx.beginPath();
0527     ctx.arc(eye_x, eye_y, L * 0.05, 0, Math.PI * 2);
0528     ctx.fill();
0529 }
0530 }
0531 }
0532 import { AcGameObject } from "./AcGameObject";
0533
0534 export class Wall extends AcGameObject {
0535     constructor(r, c, gamemap){
0536         super();
0537
0538         this.r = r;
0539         this.c = c;
0540         this.gamemap = gamemap;
0541         this.color = "#B37226";
0542     }
0543
0544     update() {
0545         this.render();
0546     }
0547
0548     render(){
0549         const L = this.gamemap.L;
0550

```

第 12 页 | 行 0551-0600 | web/src/assets/scripts/Wall.js ... web/src/components/GameMap.vue

```
0551     const ctx = this.gamemap.ctx;
0552
0553     ctx.fillStyle = this.color;
0554     ctx.fillRect(this.c * L, this.r * L, L, L);
0555   }
0556 }
0557 <template>
0558   <div class="container content-field">
0559     <div class="card">
0560       <div class="card-body">
0561         <slot></slot>
0562       </div>
0563     </div>
0564   </div>
0565 </template>
0566
0567 <script>
0568 </script>
0569
0570 <style scoped>
0571   div.content-field{
0572     margin-top: 10px;
0573   }
0574 </style>
0575 <template>
0576   <div ref="parent" class="gamemap">
0577     <canvas ref="canvas" tabindex="0"></canvas>
0578   </div>
0579 </template>
0580
0581 <script>
0582 import { GameMap } from "@assets/scripts/GameMap";
0583 import { ref, onMounted } from 'vue'
0584
0585 export default {
0586   setup() {
0587     let parent = ref(null);
0588     let canvas = ref(null);
0589
0590     onMounted(() => {
0591       new GameMap(canvas.value.getContext('2d'), parent.value)
0592     });
0593
0594     return {
0595       parent,
0596       canvas
0597     }
0598   }
0599 }
0600 </script>
```

第 13 页 | 行 0601-0650 | web/src/components/GameMap.vue ... web/src/components/MarkdownContent.vue

```

0601
0602 <style scoped>
0603   div.gamemap {
0604     width: 100%;
0605     height: 100%;
0606     display: flex;
0607     justify-content: center;
0608     align-items: center;
0609   }
0610 </style>
0611 <template>
0612   <div ref="contentRoot" class="markdown-content" v-html="html" @click="handleContentClick"></div>
0613   <Teleport to="body">
0614     <div v-if="lightboxUrl" class="markdown-lightbox" role="dialog" aria-modal="true" @click.self="closeLightbox">
0615       <button class="markdown-lightbox-close" type="button" aria-label="关闭大图" @click="closeLightbox"><\/button>
0616       
0617     <\/div>
0618   <\/Teleport>
0619 <\/template>
0620
0621 <script>
0622   import { computed, nextTick, onBeforeUnmount, ref, watch } from 'vue'
0623   import { renderMarkdown } from '../utils/markdown.mjs'
0624   import { hydrateMarkdownImages } from '../utils/markdownImages.mjs'
0625   import { lockBodyScroll, unlockBodyScroll } from '../utils/bodyScrollLock.mjs'
0626   import violationImage from '../assets/images/violation-del.png'
0627
0628   export default {
0629     name: 'MarkdownContent',
0630     props: { source: { type: String, default: '' } },
0631     setup(props) {
0632       const contentRoot = ref(null)
0633       const lightboxUrl = ref('')
0634       const lockOwner = Symbol('markdown-lightbox')
0635       let renderGeneration = 0
0636       const html = computed(() => renderMarkdown(props.source, { imagePlaceholder: violationImage }))
0637
0638       const hydrate = async () => {
0639         const generation = ++renderGeneration
0640         await nextTick()
0641         if (generation !== renderGeneration) return
0642         await hydrateMarkdownImages(contentRoot.value, () => generation === renderGeneration)
0643       }
0644       watch(() => props.source, hydrate, { immediate: true })
0645
0646       const handleContentClick = event => {
0647         const image = event.target.closest?('img')
0648         if (!image || !contentRoot.value?.contains(image)) return
0649         lightboxUrl.value = image.src
0650         lockBodyScroll(lockOwner)

```

第 14 页 | 行 0651-0700 | web/src/components/MarkdownContent.vue ... web/src/components/MarkdownEditor.vue

```

0651     }
0652     const closeLightbox = () => {
0653       lightboxUrl.value = ''
0654       unlockBodyScroll(lockOwner)
0655     }
0656     const onKeydown = event => { if (event.key === 'Escape' && lightboxUrl.value) closeLightbox() }
0657     document.addEventListener('keydown', onKeydown)
0658     onBeforeUnmount(() => {
0659       renderGeneration += 1
0660       document.removeEventListener('keydown', onKeydown)
0661       unlockBodyScroll(lockOwner)
0662     })
0663     return { contentRoot, html, lightboxUrl, handleContentClick, closeLightbox }
0664   }
0665 }
0666 </script>
0667
0668 <style>
0669 .markdown-content { line-height: 1.75; overflow-wrap: anywhere; }
0670 .markdown-content img { display: block; width: var(--markdown-image-width, 25%); max-width: 100%; height: auto; margin: 1rem auto...
0671 .markdown-content pre { padding: 1rem; overflow: auto; background: #f6f8fa; border-radius: 6px; }
0672 .markdown-content code { background: #f1f3f5; padding: .12rem .3rem; border-radius: 3px; }
0673 .markdown-content pre code { padding: 0; background: transparent; }
0674 .markdown-content blockquote { padding-left: 1rem; color: #6a737d; border-left: 4px solid #dfe2e5; }
0675 .markdown-content table { width: 100%; border-collapse: collapse; margin: 1rem 0; }
0676 .markdown-content th, .markdown-content td { padding: .5rem; border: 1px solid #dfe2e5; }
0677 .markdown-lightbox { position: fixed; inset: 0; z-index: 2000; display: flex; align-items: center; justify-content: center; paddi...
0678 .markdown-lightbox-close { position: fixed; top: 16px; right: 22px; border: 0; background: transparent; color: white; font-size: ...
0679 </style>
0680 </template>
0681 <div class="markdown-editor">
0682   <div class="markdown-editor-toolbar">
0683     <button type="button" class="btn btn-sm btn-outline-secondary" @mousedown.prevent="formatBlock('strong')">粗体</button>
0684     <button type="button" class="btn btn-sm btn-outline-secondary" @mousedown.prevent="formatBlock('h2')">标题</button>
0685     <input v-model="linkUrl" class="form-control form-control-sm markdown-link-input" type="url" placeholder="https://链接地址">
0686     <button type="button" class="btn btn-sm btn-outline-secondary" @mousedown.prevent="createLink">添加链接</button>
0687     <button type="button" class="btn btn-sm btn-outline-primary" :disabled="!blogId || uploading" @click="$refs.file.click()">
0688       {{ uploading ? '上传中...' : '上传图片' }}
0689   </button>
0690   <input ref="file" hidden type="file" accept="image/png,image/jpeg,image/webp,image/gif" @change="upload">
0691   <small v-if="!blogId" class="text-muted">先创建博客，再修改文章上传图片</small>
0692   <span v-if="error" class="text-danger">{{ error }}</span>
0693 </div>
0694 <div
0695   ref="editorRoot"
0696   class="wysiwyg-editor markdown-content"
0697   contenteditable="true"
0698   role="textbox"
0699   aria-label="博客正文"

```

第 15 页 | 行 0701-0750 | web/src/components/MarkdownEditor.vue

```

0701     aria-multiline="true"
0702     data-placeholder="请输入博客正文"
0703     @input="syncMarkdown"
0704     @click="handleEditorClick"
0705     @paste="handlePaste"
0706     @dragover.prevent
0707     @drop.prevent="handleDrop"
0708     @keydown="handleKeydown"
0709     @focus="handleFocus"
0710     @blur="handleBlur"
0711     @keyup="saveSelection"
0712     @mouseup="saveSelection"
0713     @compositionstart="composing = true"
0714     @compositionend="finishComposition"
0715   ></div>
0716 </div>
0717 </template>
0718
0719 <script>
0720 import { nextTick, onBeforeUnmount, ref, watch } from 'vue'
0721 import { useStore } from 'vuex'
0722 import { uploadBlogImage } from '../api/blogImages.mjs'
0723 import { renderMarkdown } from '../utils/markdown.mjs'
0724 import { hydrateMarkdownImages, loadImageUrl } from '../utils/markdownImages.mjs'
0725 import {
0726   adjacentImageFromSelection,
0727   calculateImageWidth,
0728   isSafeEditorUrl,
0729   sanitizeEditorFragment,
0730   serializeEditor
0731 } from '../utils/wysiwygMarkdown.mjs'
0732 import violationImage from '../assets/images/violation-del.png'
0733
0734 export default {
0735   name: 'MarkdownEditor',
0736   props: {
0737     modelValue: { type: String, default: '' },
0738     blogId: { type: [Number, String], default: null }
0739   },
0740   emits: ['update:modelValue'],
0741   setup(props, { emit }) {
0742     const store = useStore()
0743     const editorRoot = ref(null)
0744     const uploading = ref(false)
0745     const error = ref('')
0746     const composing = ref(false)
0747     const linkUrl = ref('https://')
0748     const selectedImage = ref(null)
0749     const editorHasFocus = ref(false)
0750     let emittedMarkdown = null

```

第 16 页 | 行 0751-0800 | web/src/components/MarkdownEditor.vue

```

0751     let pendingExternalMarkdown = null
0752     let savedRange = null
0753     let renderGeneration = 0
0754     let imageControls = null
0755     let resizeState = null
0756
0757     const positionImageControls = () => {
0758       if (!imageControls || !selectedImage.value || !editorRoot.value) return
0759       const editorRect = editorRoot.value.getBoundingClientRect()
0760       const imageRect = selectedImage.value.getBoundingClientRect()
0761       imageControls.style.left = `${imageRect.left - editorRect.left + editorRoot.value.scrollLeft}px`
0762       imageControls.style.top = `${imageRect.top - editorRect.top + editorRoot.value.scrollTop}px`
0763       imageControls.style.width = `${imageRect.width}px`
0764       imageControls.style.height = `${imageRect.height}px`
0765     }
0766
0767     const clearImageSelection = () => {
0768       selectedImage.value?.classList.remove('is-selected')
0769       selectedImage.value = null
0770       imageControls?.remove()
0771       imageControls = null
0772     }
0773
0774     const selectionBelongsToEditor = selection => Boolean(
0775       selection?.rangeCount && editorRoot.value?.contains(selection.anchorNode)
0776     )
0777
0778     const saveSelection = () => {
0779       const selection = window.getSelection()
0780       if (selectionBelongsToEditor(selection)) savedRange = selection.getRangeAt(0).cloneRange()
0781     }
0782
0783     const restoreSelection = range => {
0784       if (!range || !editorRoot.value?.contains(range.commonAncestorContainer)) return false
0785       const selection = window.getSelection()
0786       selection.removeAllRanges()
0787       selection.addRange(range)
0788       return true
0789     }
0790
0791     const syncMarkdown = () => {
0792       if (composing.value || !editorRoot.value) return
0793       const markdown = serializeEditor(editorRoot.value)
0794       emittedMarkdown = markdown
0795       emit('update:modelValue', markdown)
0796       saveSelection()
0797     }
0798
0799     const renderExternalMarkdown = async markdown => {
0800       const generation = ++renderGeneration

```


第 17 页 | 行 0801-0850 | web/src/components/MarkdownEditor.vue

```

0801     await nextTick()
0802     if (generation !== renderGeneration || !editorRoot.value) return
0803     clearImageSelection()
0804     editorRoot.value.innerHTML = renderMarkdown(markdown, { imagePlaceholder: violationImage })
0805     await hydrateMarkdownImages(editorRoot.value, () => generation === renderGeneration)
0806   }
0807
0808   watch(() => props.modelValue, value => {
0809     if (value === emittedMarkdown) {
0810       emittedMarkdown = null
0811       return
0812     }
0813     if (composing.value || editorHasFocus.value) {
0814       pendingExternalMarkdown = value
0815       return
0816     }
0817     pendingExternalMarkdown = null
0818     renderExternalMarkdown(value)
0819   }, { immediate: true })
0820
0821   watch(() => props.blogId, (value, previous) => {
0822     if (previous === undefined || value === previous) return
0823     emittedMarkdown = null
0824     pendingExternalMarkdown = null
0825     renderExternalMarkdown(props.modelValue)
0826   })
0827
0828   const wrapSelection = tagName => {
0829     const selection = window.getSelection()
0830     if (!selectionBelongsToEditor(selection)) return
0831     const range = selection.getRangeAt(0)
0832     const wrapper = document.createElement(tagName)
0833     if (range.collapsed) wrapper.appendChild(document.createTextNode(tagName === 'h2' ? '标题' : '文本'))
0834     else wrapper.appendChild(range.extractContents())
0835     range.insertNode(wrapper)
0836     const next = document.createRange()
0837     next.selectNodeContents(wrapper)
0838     next.collapse(false)
0839     selection.removeAllRanges()
0840     selection.addRange(next)
0841     syncMarkdown()
0842   }
0843
0844   const formatBlock = kind => wrapSelection(kind === 'strong' ? 'strong' : 'h2')
0845
0846   const createLink = () => {
0847     const href = linkUrl.value.trim()
0848     if (!isSafeEditorUrl(href)) {
0849       error.value = '请输入有效的 HTTP、HTTPS 或站内链接'
0850       return

```

第 18 页 | 行 0851-0900 | web/src/components/MarkdownEditor.vue

```

0851     }
0852     const selection = window.getSelection()
0853     if (!selectionBelongsToEditor(selection)) return
0854     const range = selection.getRangeAt(0)
0855     const link = document.createElement('a')
0856     link.setAttribute('href', href)
0857     if (range.collapsed) link.appendChild(document.createTextNode(href))
0858     else link.appendChild(range.extractContents())
0859     range.insertNode(link)
0860     error.value = ''
0861     syncMarkdown()
0862   }
0863
0864   const insertFragmentAtSelection = fragment => {
0865     const selection = window.getSelection()
0866     const range = selectionBelongsToEditor(selection) ? selection.getRangeAt(0) : null
0867     if (!range) {
0868       editorRoot.value.appendChild(fragment)
0869       return
0870     }
0871     range.deleteContents()
0872     const lastNode = fragment.lastChild
0873     range.insertNode(fragment)
0874     if (lastNode) {
0875       range.setStartAfter(lastNode)
0876       range.collapse(true)
0877       selection.removeAllRanges()
0878       selection.addRange(range)
0879     }
0880   }
0881
0882   const handlePaste = event => {
0883     event.preventDefault()
0884     const html = event.clipboardData?.getData('text/html') || ''
0885     if (html) {
0886       const template = document.createElement('template')
0887       template.innerHTML = html
0888       insertFragmentAtSelection(sanitizeEditorFragment(template.content, document))
0889     } else {
0890       const fragment = document.createDocumentFragment()
0891       const lines = (event.clipboardData?.getData('text/plain') || '').split(/\r?\n/)
0892       lines.forEach((line, index) => {
0893         if (index) fragment.appendChild(document.createElement('br'))
0894         fragment.appendChild(document.createTextNode(line))
0895       })
0896       insertFragmentAtSelection(fragment)
0897     }
0898     syncMarkdown()
0899   }
0900

```

第 19 页 | 行 0901-0950 | web/src/components/MarkdownEditor.vue

```

0901     const placeCaretAfter = node => {
0902         const selection = window.getSelection()
0903         const range = document.createRange()
0904         range.setStartAfter(node)
0905         range.collapse(true)
0906         selection.removeAllRanges()
0907         selection.addRange(range)
0908         savedRange = range.cloneRange()
0909     }
0910
0911     const insertUploadedImage = async (file, url, range, generation) => {
0912         if (!editorRoot.value) return
0913         const image = document.createElement('img')
0914         image.src = violationImage
0915         image.alt = file.name.replace(/\.(\.[^.]*)$/, '') || '图片'
0916         image.dataset.originalSrc = url
0917         image.dataset.imageWidth = '25'
0918         image.dataset.imageState = 'loading'
0919         image.style.setProperty('--markdown-image-width', '25%')
0920         if (!restoreSelection(range)) editorRoot.value.appendChild(image)
0921         else {
0922             range.deleteContents()
0923             range.insertNode(image)
0924         }
0925         if (!image.nextSibling) image.parentNode.appendChild(document.createTextNode(''))
0926         placeCaretAfter(image)
0927         syncMarkdown()
0928         try {
0929             await loadImageUrl(url)
0930             if (generation !== renderGeneration || !image.isConnected) return
0931             image.src = url
0932             image.dataset.imageState = 'loaded'
0933         } catch (_) {
0934             if (generation === renderGeneration && image.isConnected) image.dataset.imageState = 'failed'
0935         }
0936     }
0937
0938     const upload = async event => {
0939         const file = event.target.files?.[0]
0940         event.target.value = ''
0941         if (!file || !props.blogId || uploading.value) return
0942         saveSelection()
0943         const range = savedRange?.cloneRange() || null
0944         const generation = renderGeneration
0945         const blogIdAtStart = props.blogId
0946         uploading.value = true
0947         error.value = ''
0948         try {
0949             const url = await uploadBlogImage(blogIdAtStart, file, store.state.user.token)
0950             if (props.blogId !== blogIdAtStart) {

```

第 20 页 | 行 0951-1000 | web/src/components/MarkdownEditor.vue

```

0951         error.value = '文章已切换, 上传的图片未插入正文'
0952         return
0953     }
0954     await insertUploadedImage(file, url, generation === renderGeneration ? range : null, renderGeneration)
0955 } catch (uploadError) {
0956     error.value = uploadError.message || '图片上传失败'
0957 } finally {
0958     uploading.value = false
0959 }
0960 }
0961
0962 const continueResize = event => {
0963     if (!resizeState || event.pointerId !== resizeState.pointerId) return
0964     const distance = Math.abs(event.clientX - resizeState.centerX) * 2
0965     const virtualPointerX = resizeState.editorLeft + distance
0966     const width = calculateImageWidth(virtualPointerX, resizeState.editorLeft, resizeState.editorWidth)
0967     resizeState.image.dataset.imageWidth = String(width)
0968     resizeState.image.style.setProperty('--markdown-image-width', `${width}%`)
0969     positionImageControls()
0970 }
0971
0972 const removeResizeListeners = () => {
0973     window.removeEventListener('pointermove', continueResize)
0974     window.removeEventListener('pointerup', finishResize)
0975     window.removeEventListener('pointercancel', finishResize)
0976 }
0977
0978 const finishResize = event => {
0979     if (!resizeState || (event?.pointerId !== null && event.pointerId !== resizeState.pointerId)) return
0980     removeResizeListeners()
0981     resizeState = null
0982     positionImageControls()
0983     syncMarkdown()
0984 }
0985
0986 const beginResize = (event, side) => {
0987     if (!selectedImage.value || !editorRoot.value) return
0988     event.preventDefault()
0989     event.stopPropagation()
0990     const editorRect = editorRoot.value.getBoundingClientRect()
0991     resizeState = {
0992         image: selectedImage.value,
0993         side,
0994         pointerId: event.pointerId,
0995         editorLeft: editorRect.left,
0996         editorWidth: editorRect.width,
0997         centerX: editorRect.left + editorRect.width / 2
0998     }
0999     window.addEventListener('pointermove', continueResize)
1000     window.addEventListener('pointerup', finishResize)

```

第 21 页 | 行 1001-1050 | web/src/components/MarkdownEditor.vue

```

1001     window.addEventListener('pointercancel', finishResize)
1002   }
1003
1004   const createResizeHandle = (side, label) => {
1005     const handle = document.createElement('button')
1006     handle.type = 'button'
1007     handle.className = `image-resize-handle image-resize-handle-${side}`
1008     handle.dataset.editorUi = 'resize-handle'
1009     handle.contentEditable = 'false'
1010     handle.setAttribute('aria-label', label)
1011     handle.addEventListener('pointerdown', event => beginResize(event, side))
1012     handle.addEventListener('click', event => event.stopPropagation())
1013     return handle
1014   }
1015
1016   const selectImage = image => {
1017     clearImageSelection()
1018     selectedImage.value = image
1019     image.classList.add('is-selected')
1020     imageControls = document.createElement('span')
1021     imageControls.className = 'image-resize-controls'
1022     imageControls.dataset.editorUi = 'image-controls'
1023     imageControls.contentEditable = 'false'
1024     imageControls.appendChild(createResizeHandle('left', '从左侧调整图片大小'))
1025     imageControls.appendChild(createResizeHandle('right', '从右侧调整图片大小'))
1026     editorRoot.value.appendChild(imageControls)
1027     positionImageControls()
1028   }
1029
1030   const handleEditorClick = event => {
1031     if (event.target.closest?(['[data-editor-ui]'])) return
1032     const image = event.target.closest?(['img[data-original-src]'])
1033     if (image && editorRoot.value?.contains(image)) {
1034       event.preventDefault()
1035       selectImage(image)
1036     } else clearImageSelection()
1037   }
1038
1039   const placeCaretAtRemovalPoint = (parent, index) => {
1040     if (!parent || !editorRoot.value?.contains(parent) && parent !== editorRoot.value) return
1041     const range = document.createRange()
1042     range.setStart(parent, Math.min(index, parent.childNodes.length))
1043     range.collapse(true)
1044     const selection = window.getSelection()
1045     selection.removeAllRanges()
1046     selection.addRange(range)
1047     savedRange = range.cloneRange()
1048   }
1049
1050   const removeImageFromEditor = image => {

```

第 22 页 | 行 1051-1100 | web/src/components/MarkdownEditor.vue

```

1051     const parent = image.parentNode
1052     const index = [...parent.childNodes].indexOf(image)
1053     clearImageSelection()
1054     image.remove()
1055     placeCaretAtRemovalPoint(parent, index)
1056     syncMarkdown()
1057   }
1058
1059   const handleKeydown = event => {
1060     if (event.key === 'Escape') {
1061       clearImageSelection()
1062       savedRange = null
1063       return
1064     }
1065     const deletingBackward = event.key === 'Backspace'
1066     const deletingForward = event.key === 'Delete'
1067     if (!deletingBackward && !deletingForward) {
1068       clearImageSelection()
1069       return
1070     }
1071     const selection = window.getSelection()
1072     const image = selectedImage.value || adjacentImageFromSelection(
1073       selection,
1074       deletingBackward ? 'backward' : 'forward'
1075     )
1076     if (!image || !editorRoot.value?.contains(image)) return
1077     event.preventDefault()
1078     removeImageFromEditor(image)
1079   }
1080
1081   const finishComposition = () => {
1082     composing.value = false
1083     syncMarkdown()
1084   }
1085
1086   const handleFocus = () => {
1087     editorHasFocus.value = true
1088     saveSelection()
1089   }
1090
1091   const handleBlur = () => {
1092     editorHasFocus.value = false
1093     if (pendingExternalMarkdown === null || !editorRoot.value) return
1094     const value = props.modelValue
1095     pendingExternalMarkdown = null
1096     if (value !== serializeEditor(editorRoot.value)) renderExternalMarkdown(value)
1097   }
1098
1099   const handleDrop = () => {
1100     error.value = '请使用“上传图片”按钮添加本地图片'

```

第 23 页 | 行 1101-1150 | web/src/components/MarkdownEditor.vue

```

1101     }
1102
1103     const onSelectionChange = () => saveSelection()
1104     document.addEventListener('selectionchange', onSelectionChange)
1105     onBeforeUnmount(() => {
1106       renderGeneration += 1
1107       removeResizeListeners()
1108       clearImageSelection()
1109       document.removeEventListener('selectionchange', onSelectionChange)
1110     })
1111
1112     return {
1113       editorRoot,
1114       uploading,
1115       error,
1116       composing,
1117       editorHasFocus,
1118       linkUrl,
1119       formatBlock,
1120       createLink,
1121       syncMarkdown,
1122       handlePaste,
1123       handleDrop,
1124       handleEditorClick,
1125       handleKeydown,
1126       beginResize,
1127       continueResize,
1128       finishResize,
1129       finishComposition,
1130       handleFocus,
1131       handleBlur,
1132       saveSelection,
1133       upload
1134     }
1135   }
1136 }
1137 </script>
1138
1139 <style scoped>
1140 .markdown-editor-toolbar { display: flex; align-items: center; flex-wrap: wrap; gap: .5rem; margin-bottom: .75rem; }
1141 .markdown-link-input { width: min(260px, 100%); }
1142 .wysiwyg-editor { min-height: 350px; padding: 1rem; overflow-y: auto; border: 1px solid #dee2e6; border-radius: 4px; background: ...
1143 .wysiwyg-editor { position: relative; }
1144 .wysiwyg-editor:focus { border-color: #86b7fe; box-shadow: 0 0 0 .2rem rgba(13, 110, 253, .15); }
1145 .wysiwyg-editor:empty::before { color: #adb5bd; content: attr(data-placeholder); pointer-events: none; }
1146 .wysiwyg-editor :deep(img) { display: block; width: var(--markdown-image-width, 25%); max-width: 100%; height: auto; margin: 1rem...
1147 .wysiwyg-editor :deep(img.is-selected) { outline: 3px solid #0d6efd; outline-offset: 3px; }
1148 .wysiwyg-editor :deep(.image-resize-controls) { position: absolute; z-index: 2; display: block; pointer-events: none; }
1149 .wysiwyg-editor :deep(.image-resize-handle) { position: absolute; top: 50%; width: 16px; height: 32px; padding: 0; border: 2px so...
1150 .wysiwyg-editor :deep(.image-resize-handle-left) { left: -8px; }

```

第 24 页 | 行 1151-1200 | web/src/components/MarkdownEditor.vue ... web/src/components/NavBar.vue

```

1151 .wysiwyg-editor :deep(.image-resize-handle-right) { right: -8px; }
1152 </style>
1153 <template>
1154 <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
1155   <div class="container">
1156     <router-link class="navbar-brand" :to="{name:'home'}">Blog and Oj</router-link>
1157     <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarText" aria-controls="navbarText...
1158     <span class="navbar-toggler-icon"></span>
1159   </button>
1160   <div class="collapse navbar-collapse" id="navbarText">
1161     <ul class="navbar-nav me-auto mb-4 mb-lg-0">
1162       <li class="nav-item">
1163         <router-link :class="route_name == 'pk_index' ? 'nav-link active' : 'nav-link'" :to="{name:'pk_index'}">博
客</router-...
1164       </li>
1165       <li class="nav-item">
1166         <router-link :class="route_name == 'Ranklist_index' ? 'nav-link active' : 'nav-link'" :to="{name:'Ranklist_index'}"...
1167       </li>
1168       <li class="nav-item dropdown" v-if="admin">
1169         <a
1170           class="nav-link dropdown-toggle"
1171           :class="route_name && route_name.toString().startsWith('manage_') ? 'active' : ''"
1172           href="#"
1173           role="button"
1174           data-bs-toggle="dropdown"
1175           aria-expanded="false"
1176         >
1177           后台管理
1178         </a>
1179         <ul class="dropdown-menu">
1180           <li><router-link class="dropdown-item" :to="{name:'manage_users'}">用户管理</router-link></li>
1181           <li><router-link class="dropdown-item" :to="{name:'manage_login_records'}">用户登录信息</router-link></li>
1182           <li><router-link class="dropdown-item" :to="{name:'manage_topics'}">题库管理</router-link></li>
1183         </ul>
1184       </li>
1185       <!-- <li class="nav-item">
1186         <router-link :class="route_name == 'record_index' ? 'nav-link active' : 'nav-link'" :to="{name:'record_index'}">排行
榜...
1187       </li> -->
1188     </ul>
1189     <ul class="navbar-nav" v-if="$store.state.user.is_login">
1190       <li class="nav-item dropdown">
1191         <a class="nav-link dropdown-toggle" href="#" role="button" data-bs-toggle="dropdown" aria-expanded="false">
1192           {{ $store.state.user.username }}
1193         </a>
1194         <ul class="dropdown-menu">
1195           <li>
1196             <router-link class="dropdown-item" :to="{name:'userbot_index'}">个人空间</router-link>
1197           </li>
1198           <li>
1199             <router-link class="dropdown-item" :to="{name:'settings_index'}">设置</router-link>
1200           </li>

```


第25页 | 行 1201-1250 | web/src/components/NavBar.vue

```

1201         <li><hr class="dropdown-divider"></li>
1202         <li>
1203             <a class="dropdown-item" href="#" @click="logout">退出</a>
1204         </li>
1205     </ul>
1206 </li>
1207 </ul>
1208 <ul class="navbar-nav" v-else-if="!$store.state.user.pulling_info">
1209     <li class="nav-item">
1210         <router-link class="nav-link" :to="{name: 'user_account_login'}" role="button">
1211             登录
1212         </router-link>
1213     </li>
1214     <li class="nav-item">
1215         <router-link class="nav-link" :to="{name: 'user_account_register'}" role="button">
1216             注册
1217         </router-link>
1218     </li>
1219 </ul>
1220 </div>
1221 </div>
1222 </nav>
1223 </template>
1224
1225 <script>
1226 import { useRoute } from 'vue-router';
1227 import { computed } from 'vue';
1228 import { useStore } from 'vuex';
1229 import { isAdmin } from '@utils/admin.mjs';
1230
1231 export default{
1232     setup(){
1233         const store = useStore();
1234         const route = useRoute();
1235         let route_name = computed(() => route.name)
1236         const admin = computed(() => isAdmin(store.state.user.root))
1237
1238         const logout = () => {
1239             store.dispatch("logout");
1240         }
1241
1242         return{
1243             route_name,
1244             admin,
1245             logout,
1246         }
1247     }
1248 }
1249 </Script>
1250

```

第 26 页 | 行 1251-1300 | web/src/components/NavBar.vue ... web/src/components/ResultModal.vue

```

1251 <style scoped>
1252
1253 </style>
1254 <template>
1255   <div class = "playground">
1256     <GameMap />
1257   </div>
1258 </template>
1259
1260 <script>
1261 import GameMap from './GameMap.vue';
1262
1263 export default{
1264   components:{
1265     GameMap
1266   }
1267 }
1268 </script>
1269
1270 <style scoped>
1271
1272 div.playground{
1273   width: 60vw;
1274   height: 70vh;
1275   margin: 60px auto;
1276 }
1277 </style>
1278 <template>
1279   <div class="modal-overlay" v-if="visible" @click.self="closeModal">
1280     <div class="modal-container" :class="resultClass">
1281       <div class="modal-header">
1282         <div class="result-icon">
1283           <i v-if="resultData.state === 'accepted'" class="bi bi-check-circle"></i>
1284           <i v-else-if="resultData.state === 'wrong_answer'" class="bi bi-x-circle"></i>
1285           <i v-else-if="resultData.state === 'time_limit'" class="bi bi-clock"></i>
1286           <i v-else-if="resultData.state === 'runtime_error'" class="bi bi-exclamation-triangle"></i>
1287           <i v-else class="bi bi-info-circle"></i>
1288         </div>
1289       </div>
1290
1291       <div class="modal-body">
1292         <h3 class="result-title">{{ resultTitle }}</h3>
1293         <p class="result-desc">{{ resultDesc }}</p>
1294
1295         <div v-if="!isSubmissionError" class="result-score">
1296           <div class="score-circle">
1297             <span class="score-value">{{ resultData.score }}</span>
1298             <span class="score-label">分</span>
1299           </div>
1300         </div>

```

第 27 页 | 行 1301-1350 | web/src/components/ResultModal.vue

```

1301
1302     <div v-if="!isSubmissionError" class="result-details">
1303       <div class="detail-item">
1304         <span class="detail-label">题目 ID</span>
1305         <span class="detail-value">#{ resultData.evaluation_id }</span>
1306       </div>
1307       <div class="detail-item">
1308         <span class="detail-label">状态</span>
1309         <span class="detail-value">{{ stateText }}</span>
1310       </div>
1311     </div>
1312   </div>
1313
1314   <div class="modal-footer">
1315     <button class="btn btn-primary" @click="closeModal">
1316       {{ isSubmissionError ? '返回修改' : (resultData.state === 'accepted' ? '继续挑战' : '修改代码') }}
1317     </button>
1318     <button v-if="resultData.state !== 'accepted' && !isSubmissionError" class="btn btn-outline-secondary" @click="viewSoluti...
1319       查看题解
1320     </button>
1321   </div>
1322 </div>
1323 </div>
1324 </template>
1325
1326 <script>
1327 import { computed } from 'vue'
1328
1329 export default {
1330   name: 'ResultModal',
1331
1332   props: {
1333     visible: {
1334       type: Boolean,
1335       default: false
1336     },
1337     resultData: {
1338       type: Object,
1339       default: () => ({
1340         user_id: null,
1341         evaluation_id: null,
1342         score: null,
1343         state: '',
1344         message: ''
1345       })
1346     }
1347   },
1348
1349   emits: ['close', 'viewSolution'],
1350

```

第 28 页 | 行 1351-1400 | web/src/components/ResultModal.vue

```

1351 setup(props, { emit }) {
1352   const closeModal = () => {
1353     emit('close')
1354   }
1355
1356   const viewSolution = () => {
1357     emit('viewSolution', props.resultData.evaluation_id)
1358   }
1359
1360   const isSubmissionError = computed(() => props.resultData.state === 'submission_error')
1361
1362   const resultClass = computed(() => {
1363     const classes = {
1364       'result-success': props.resultData.state === 'accepted',
1365       'result-warning': props.resultData.state === 'wrong_answer',
1366       'result-danger': ['time_limit', 'runtime_error', 'submission_error'].includes(props.resultData.state),
1367       'result-info': ![ 'accepted', 'wrong_answer', 'time_limit', 'runtime_error', 'submission_error' ].includes(props.resultData...)
1368     }
1369     return classes
1370   })
1371
1372   const resultTitle = computed(() => {
1373     const titles = {
1374       accepted: '恭喜通过! ',
1375       wrong_answer: '答案错误',
1376       time_limit: '超时',
1377       runtime_error: '运行错误',
1378       compile_error: '编译错误',
1379       submission_error: '提交失败'
1380     }
1381     return titles[props.resultData.state] || '判题完成'
1382   })
1383
1384   const resultDesc = computed(() => {
1385     if (isSubmissionError.value) {
1386       return props.resultData.message || '提交失败, 请稍后重试'
1387     }
1388
1389     const desc = {
1390       accepted: '你的代码通过了所有测试用例, 表现出色! ',
1391       wrong_answer: '你的代码输出与预期不符, 请检查逻辑',
1392       time_limit: '代码运行时间过长, 尝试优化算法',
1393       runtime_error: '程序运行时发生错误, 请检查代码',
1394       compile_error: '代码编译失败, 请检查语法错误'
1395     }
1396     return desc[props.resultData.state] || '判题已完成'
1397   })
1398
1399   const stateText = computed(() => {
1400     const texts = {

```

第29页 | 行 1401-1450 | web/src/components/ResultModal.vue

```
1401         accepted: '通过',
1402         wrong_answer: '答案错误',
1403         time_limit: '超时',
1404         runtime_error: '运行错误',
1405         compile_error: '编译错误'
1406     }
1407     return texts[props.resultData.state] || props.resultData.state
1408 })
1409
1410     return {
1411         closeModal,
1412         viewSolution,
1413         isSubmissionError,
1414         resultClass,
1415         resultTitle,
1416         resultDesc,
1417         stateText
1418     }
1419 }
1420 }
1421 </script>
1422
1423 <style scoped>
1424 .modal-overlay {
1425     position: fixed;
1426     top: 0;
1427     left: 0;
1428     right: 0;
1429     bottom: 0;
1430     background-color: rgba(0, 0, 0, 0.5);
1431     display: flex;
1432     align-items: center;
1433     justify-content: center;
1434     z-index: 1000;
1435     animation: fadeIn 0.3s ease;
1436 }
1437
1438 @keyframes fadeIn {
1439     from {
1440         opacity: 0;
1441     }
1442     to {
1443         opacity: 1;
1444     }
1445 }
1446
1447 .modal-container {
1448     background: white;
1449     border-radius: 20px;
1450     padding: 2rem;
```

第 30 页 | 行 1451-1500 | web/src/components/ResultModal.vue

```
1451   min-width: 400px;
1452   max-width: 90%;
1453   text-align: center;
1454   animation: slideUp 0.4s ease;
1455   box-shadow: 0 20px 60px rgba(0, 0, 0, 0.3);
1456 }
1457
1458 @keyframes slideUp {
1459   from {
1460     opacity: 0;
1461     transform: translateY(30px);
1462   }
1463   to {
1464     opacity: 1;
1465     transform: translateY(0);
1466   }
1467 }
1468
1469 .modal-container.result-success {
1470   border-top: 6px solid #28a745;
1471 }
1472
1473 .modal-container.result-warning {
1474   border-top: 6px solid #ffc107;
1475 }
1476
1477 .modal-container.result-danger {
1478   border-top: 6px solid #dc3545;
1479 }
1480
1481 .modal-container.result-info {
1482   border-top: 6px solid #17a2b8;
1483 }
1484
1485 .modal-header {
1486   margin-bottom: 1.5rem;
1487 }
1488
1489 .result-icon {
1490   font-size: 6rem;
1491 }
1492
1493 .result-success .result-icon i {
1494   color: #28a745;
1495 }
1496
1497 .result-warning .result-icon i {
1498   color: #ffc107;
1499 }
1500
```

第31页 | 行 1501-1550 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/evaluate/EndEvaluateServiceImpl.java

```
1501 import com.alibaba.fastjson2.JSONObject;
1502 import com.kob.backend.consumer.WebSocketServer;
1503 import com.kob.backend.mapper.RecordOfQuestionMapper;
1504 import com.kob.backend.mapper.TopicMapper;
1505 import com.kob.backend.pojo.RecordOfQuestion;
1506 import com.kob.backend.service.oj.evaluate.EndEvaluateService;
1507 import lombok.Data;
1508 import org.springframework.amqp.rabbit.annotation.RabbitListener;
1509 import org.springframework.amqp.rabbit.core.RabbitTemplate;
1510 import org.springframework.beans.factory.annotation.Autowired;
1511 import org.springframework.boot.context.event.ApplicationReadyEvent;
1512 import org.springframework.context.event.EventListener;
1513 import org.springframework.stereotype.Component;
1514 import org.springframework.stereotype.Service;
1515
1516 import javax.sql.DataSource;
1517 import java.util.Date;
1518 import java.util.concurrent.BlockingQueue;
1519 import java.util.concurrent.LinkedBlockingQueue;
1520
1521 @Component
1522 public class EndEvaluateServiceImpl extends Thread {
1523     @Autowired
1524     private RecordOfQuestionMapper recordOfQuestionMapper;
1525     @Autowired
1526     private TopicMapper topicMapper;
1527     @Autowired
1528     RabbitTemplate rabbitTemplate;
1529
1530     BlockingQueue<JSONObject> queue = new LinkedBlockingQueue<>();
1531
1532     @RabbitListener (queues = "evaluate.result.queue")
1533     public void EvaluateResult(JSONObject message) {
1534         try {
1535             queue.put(message);
1536         } catch (InterruptedException e) {
1537             throw new RuntimeException(e);
1538         }
1539     }
1540
1541     @EventListener(ApplicationReadyEvent.class)
1542     public void startThread(){
1543         this.start();
1544     }
1545
1546     @Override
1547     public void run() {
1548         while (true) {
1549             JSONObject message;
1550             try {
```

第 32 页 | 行 1551-1600 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/evaluate/EndEvaluateServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/record/AddRecordRecordServiceImpl.java

```

1551         message = queue.take();
1552     } catch (InterruptedException e) {
1553         throw new RuntimeException(e);
1554     }
1555     System.out.println("End: " + message);
1556     EndEvaluate(message);
1557 }
1558 }
1559
1560 public String EndEvaluate(JSONObject jsonObject) {
1561     System.out.println(jsonObject.getString("score"));
1562     Integer user_id = Integer.parseInt(jsonObject.getString("user_id"));
1563     Integer evaluate_id = Integer.parseInt(jsonObject.getString("evaluate_id"));
1564     Integer score = Integer.parseInt(jsonObject.getString("score"));
1565     String state = jsonObject.getString("state");
1566
1567     String title = topicMapper.selectById(evaluate_id).getTitle();
1568     Date date = new Date();
1569     RecordOfQuestion recordOfQuestion = new RecordOfQuestion(null,
1570         user_id,
1571         evaluate_id,
1572         title,
1573         state,
1574         score,
1575         date
1576     );
1577     recordOfQuestionMapper.insert(recordOfQuestion);
1578     WebSocketServer.startEvaluate(user_id, evaluate_id, score, state);
1579     return "success";
1580 }
1581 }
1582 package com.kob.backend.service.impl.oj.record;
1583
1584
1585 import com.kob.backend.mapper.RecordOfQuestionMapper;
1586 import com.kob.backend.pojo.RecordOfQuestion;
1587 import com.kob.backend.pojo.User;
1588 import com.kob.backend.service.impl.utils.UserDetailsImpl;
1589 import com.kob.backend.service.oj.record.AddRecordService;
1590 import org.springframework.beans.factory.annotation.Autowired;
1591 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
1592 import org.springframework.security.core.context.SecurityContextHolder;
1593 import org.springframework.stereotype.Service;
1594
1595 import java.util.Date;
1596 import java.util.HashMap;
1597 import java.util.Map;
1598
1599 @Service
1600 public class AddRecordRecordServiceImpl implements AddRecordService {

```


第33页 | 行 1601-1650 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/record/AddRecordRecordServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/record/GetListRecordRecordServiceImpl.java

```

1601     @Autowired
1602     private RecordOfQuestionMapper recordOfQuestionMapper;
1603
1604     @Override
1605     public Map<String, String> add(Map<String, String> data) {
1606         UsernamePasswordAuthenticationToken authenticationToken =
1607             (UsernamePasswordAuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
1608         UserDetailsImpl loginUser = (UserDetailsImpl) authenticationToken.getPrincipal();
1609         User user = loginUser.getUser();
1610
1611         Map<String, String> map = new HashMap<>();
1612
1613         int question_id = Integer.parseInt(data.get("question_id"));
1614         String title = data.get("title");
1615         String state = data.get("state");
1616         int score = Integer.parseInt(data.get("score"));
1617
1618         Date date = new Date();
1619         RecordOfQuestion recordOfQuestion = new RecordOfQuestion(null, user.getId(), question_id, title, state, score, date);
1620         recordOfQuestionMapper.insert(recordOfQuestion);
1621         map.put("error_message", "success");
1622
1623         return map;
1624     }
1625 }
1626 package com.kob.backend.service.impl.oj.record;
1627
1628 import com.baomidou.mybatisplus.core.conditions.query.QueryWrapper;
1629 import com.kob.backend.mapper.RecordOfQuestionMapper;
1630 import com.kob.backend.pojo.RecordOfQuestion;
1631 import com.kob.backend.pojo.User;
1632 import com.kob.backend.service.impl.utils.UserDetailsImpl;
1633 import com.kob.backend.service.oj.record.GetListRecordService;
1634 import org.springframework.beans.factory.annotation.Autowired;
1635 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
1636 import org.springframework.security.core.context.SecurityContextHolder;
1637 import org.springframework.stereotype.Service;
1638
1639 import java.util.List;
1640
1641 @Service
1642 public class GetListRecordRecordServiceImpl implements GetListRecordService {
1643     @Autowired
1644     RecordOfQuestionMapper recordOfQuestionMapper;
1645
1646     @Override
1647     public List<RecordOfQuestion> getList() {
1648         UsernamePasswordAuthenticationToken authenticationToken =
1649             (UsernamePasswordAuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
1650         UserDetailsImpl loginUser = (UserDetailsImpl) authenticationToken.getPrincipal();

```

第 34 页 | 行 1651-1700 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/record/GetListRecordRecordServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/AddTopicServiceImpl.java

```

1651         User user = loginUser.getUser();
1652
1653         QueryWrapper<RecordOfQuestion> queryWrapper = new QueryWrapper<>();
1654         queryWrapper.eq("user_id",user.getId());
1655
1656         return recordOfQuestionMapper.selectList(queryWrapper);
1657     }
1658 }
1659 package com.kob.backend.service.impl.oj.topic;
1660
1661 import com.baomidou.mybatisplus.core.conditions.query.QueryWrapper;
1662 import com.kob.backend.mapper.TopicMapper;
1663 import com.kob.backend.pojo.Topic;
1664 import com.kob.backend.pojo.User;
1665 import com.kob.backend.service.impl.utils.UserDetailsImpl;
1666 import com.kob.backend.service.oj.topic.AddTopicService;
1667 import com.kob.backend.utils.BloomInitializer;
1668 import org.springframework.beans.factory.annotation.Autowired;
1669 import org.springframework.beans.factory.annotation.Qualifier;
1670 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
1671 import org.springframework.security.core.context.SecurityContextHolder;
1672 import org.springframework.stereotype.Service;
1673
1674 import java.util.HashMap;
1675 import java.util.Map;
1676
1677 @Service
1678 public class AddTopicServiceImpl implements AddTopicService {
1679     @Autowired
1680     private TopicMapper topicMapper;
1681
1682     @Autowired
1683     private BloomInitializer bloomInitializer;
1684     @Override
1685     public Map<String, String> add(Map<String, String> data) {
1686         UsernamePasswordAuthenticationToken authenticationToken =
1687             (UsernamePasswordAuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
1688         UserDetailsImpl loginUser = (UserDetailsImpl) authenticationToken.getPrincipal();
1689         User user = loginUser.getUser();
1690         Map<String,String> map=new HashMap<>();
1691
1692         if(!user.getRole().equals("true")){
1693             map.put("error_message", "你没有权限进行此操作");
1694             return map;
1695         }
1696
1697         if(data.get("test_point")==null || data.get("test_point").equals("0")){
1698             map.put("error_message", "测试点数量不能为 0");
1699             return map;
1700         }

```

第 35 页 | 行 1701-1750 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/AddTopicServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/GetListTopicServiceImpl.java

```

1701         if(data.get("title")==null || data.get("title").isEmpty()){
1702             map.put("error_message", "标题不能为空");
1703             return map;
1704         }
1705         if(data.get("title").length()>100){
1706             map.put("error_message", "标题的长度不能超过 100");
1707             return map;
1708         }
1709         if(data.get("description")==null || data.get("description").isEmpty()){
1710             map.put("error_message", "题目描述不能为空");
1711             return map;
1712         }
1713         if(data.get("time_limit") ==null || data.get("time_limit").equals("0")){
1714             map.put("error_message", "时间限制不能为 0");
1715             return map;
1716         }
1717         if(data.get("mem_limit") ==null || data.get("mem_limit").equals("0")){
1718             map.put("error_message", "内存限制不能为 0");
1719             return map;
1720         }
1721         Topic topic=new Topic(null,
1722             Integer.parseInt(data.get("test_point")),
1723             data.get("title"),
1724             data.get("description"),
1725             data.get("star"),
1726             Integer.parseInt(data.get("time_limit")),
1727             Integer.parseInt(data.get("mem_limit")),
1728             data.get("input_format"),
1729             data.get("output_format"),
1730             data.get("sample_input"),
1731             data.get("sample_output"),
1732             data.get("hint")
1733         );
1734         topicMapper.insert(topic);
1735         bloomInitializer.getBloomFilter_Topic().put(topic.getId());
1736         map.put("error_message", "success");
1737         return map;
1738     }
1739 }
1740 package com.kob.backend.service.impl.oj.topic;
1741
1742 import com.baomidou.mybatisplus.core.conditions.query.LambdaQueryWrapper;
1743 import com.baomidou.mybatisplus.extension.plugins.pagination.Page;
1744 import com.kob.backend.mapper.TopicMapper;
1745 import com.kob.backend.model.PageResponse;
1746 import com.kob.backend.pojo.Topic;
1747 import com.kob.backend.service.oj.topic.GetListTopicService;
1748 import com.kob.backend.utils.PaginationUtils;
1749 import org.springframework.stereotype.Service;
1750

```

第36页 | 行 1751-1800 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/GetListTopicServiceImpl.java

```

1751 import java.util.List;
1752
1753 @Service
1754 public class GetListTopicServiceImpl implements GetListTopicService {
1755     private static final long PAGE_SIZE = 20L;
1756
1757     private final TopicMapper topicMapper;
1758
1759     public GetListTopicServiceImpl(TopicMapper topicMapper) {
1760         this.topicMapper = topicMapper;
1761     }
1762
1763     @Override
1764     public List<Topic> getList() {
1765         return topicMapper.selectList(null);
1766     }
1767
1768     @Override
1769     public PageResponse<Topic> getList(long page, String keyword) {
1770         long requestedPage = Math.max(1L, page);
1771         String normalizedKeyword = PaginationUtils.normalizeKeyword(keyword);
1772         LambdaQueryWrapper<Topic> query = new LambdaQueryWrapper<>();
1773         if (!normalizedKeyword.isEmpty()) {
1774             Integer topicId = parseTopicId(normalizedKeyword);
1775             query.and(item -> {
1776                 if (topicId != null) {
1777                     item.eq(Topic::getId, topicId).or();
1778                 }
1779                 item.like(Topic::getTitle, normalizedKeyword)
1780                     .or()
1781                     .like(Topic::getDescription, normalizedKeyword);
1782             });
1783         }
1784         query.orderByAsc(Topic::getId);
1785
1786         Page<Topic> result = topicMapper.selectPage(new Page<>(requestedPage, PAGE_SIZE), query);
1787         long total = result.getTotal();
1788         long totalPages = result.getPages();
1789         long currentPage = totalPages == 0L ? 1L : Math.min(requestedPage, totalPages);
1790         if (currentPage != requestedPage) {
1791             result = topicMapper.selectPage(new Page<>(currentPage, PAGE_SIZE, false), query);
1792         }
1793
1794         return new PageResponse<>(result.getRecords(), currentPage, PAGE_SIZE, total, totalPages);
1795     }
1796
1797     private Integer parseTopicId(String keyword) {
1798         if (!keyword.matches("\\d+")) {
1799             return null;
1800         }

```

第 37 页 | 行 1801-1850 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/GetListTopicServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/GetTopicServiceImpl.java

```

1801         try {
1802             return Integer.valueOf(keyword);
1803         } catch (NumberFormatException exception) {
1804             return null;
1805         }
1806     }
1807 }
1808 package com.kob.backend.service.impl.oj.topic;
1809
1810 import com.fasterxml.jackson.core.JsonProcessingException;
1811 import com.fasterxml.jackson.databind.ObjectMapper;
1812 import com.kob.backend.mapper.TopicMapper;
1813 import com.kob.backend.pojo.Topic;
1814 import com.kob.backend.service.oj.topic.GetTopicService;
1815 import com.kob.backend.utils.BloomInitializer;
1816 import org.springframework.beans.factory.annotation.Autowired;
1817 import org.springframework.data.redis.core.StringRedisTemplate;
1818 import org.springframework.stereotype.Service;
1819
1820 import java.util.HashMap;
1821 import java.util.Map;
1822
1823 @Service
1824 public class GetTopicServiceImpl implements GetTopicService {
1825     @Autowired
1826     private TopicMapper topicMapper;
1827
1828     @Autowired
1829     private StringRedisTemplate redisTemplate;
1830
1831     @Autowired
1832     private ObjectMapper objectMapper;
1833
1834     @Autowired
1835     private BloomInitializer bloomInitializer;
1836
1837     @Override
1838     public Map<String, String> getTopic(Integer topic_id) throws JsonProcessingException {
1839         Topic topic = new Topic();
1840         Map<String, String> map = new HashMap<>();
1841         if(!bloomInitializer.getBloomFilter_Topic().mightContain(topic_id)){
1842             map.put("error_message", "该题目不存在或已被删除");
1843             return map;
1844         }
1845         if(!redisTemplate.hasKey("topic:" + topic_id)){
1846             topic = topicMapper.selectById(topic_id);
1847             if(topic == null){
1848                 map.put("error_message", "该题目不存在或已被删除");
1849                 return map;
1850             }

```

第 38 页 | 行 1851-1900 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/GetTopicServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/RemoveTopicServiceImpl.java

```

1851         String topic_string = objectMapper.writeValueAsString(topic);
1852         redisTemplate.opsForValue().set("topic:" + topic_id, topic_string);
1853     }
1854     else{
1855         String topic_string = redisTemplate.opsForValue().get("topic:" + topic_id);
1856         topic = objectMapper.readValue(topic_string, Topic.class);
1857     }
1858     map.put("test_point", topic.getTestPoint().toString());
1859     map.put("title", topic.getTitle());
1860     map.put("description", topic.getDescription());
1861     map.put("star", topic.getStar());
1862     map.put("time_limit", topic.getTimeLimit().toString());
1863     map.put("mem_limit", topic.getMemLimit().toString());
1864     map.put("input_format", topic.getInputFormat());
1865     map.put("output_format", topic.getOutputFormat());
1866     map.put("sample_input", topic.getSampleInput());
1867     map.put("sample_output", topic.getSampleOutput());
1868     map.put("hint", topic.getHint());
1869     return map;
1870 }
1871 }
1872 package com.kob.backend.service.impl.oj.topic;
1873
1874 import com.kob.backend.mapper.TopicMapper;
1875 import com.kob.backend.pojo.Topic;
1876 import com.kob.backend.pojo.User;
1877 import com.kob.backend.service.impl.utils.UserDetailsImpl;
1878 import com.kob.backend.service.oj.topic.RemoveTopicService;
1879 import org.springframework.beans.factory.annotation.Autowired;
1880 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
1881 import org.springframework.security.core.context.SecurityContextHolder;
1882 import org.springframework.stereotype.Service;
1883
1884 import java.util.HashMap;
1885 import java.util.Map;
1886
1887 @Service
1888 public class RemoveTopicServiceImpl implements RemoveTopicService {
1889     @Autowired
1890     TopicMapper topicMapper;
1891
1892     @Override
1893     public Map<String, String> removeTopic(Map<String, String> data) {
1894         UsernamePasswordAuthenticationToken authenticationToken =
1895             (UsernamePasswordAuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
1896         UserDetailsImpl loginUser = (UserDetailsImpl) authenticationToken.getPrincipal();
1897         User user = loginUser.getUser();
1898
1899         Map<String, String> map = new HashMap<>();
1900         Integer topic_id = Integer.parseInt(data.get("topic_id"));

```

第39页 | 行 1901-1950 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/RemoveTopicServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/UdataTopicServiceImpl.java

```

1901
1902     Topic topic = topicMapper.selectById(topic_id);
1903     if(!user.getRoot().equals("true")){
1904         map.put("error_message", "你没有权限删除该题目");
1905         return map;
1906     } else if (topic==null) {
1907         map.put("error_message", "该题目不存在或已被删除");
1908     }
1909
1910     topicMapper.deleteById(topic_id);
1911     map.put("error_message", "success");
1912     return map;
1913 }
1914 }
1915 package com.kob.backend.service.impl.oj.topic;
1916
1917 import com.kob.backend.mapper.TopicMapper;
1918 import com.kob.backend.pojo.Topic;
1919 import com.kob.backend.pojo.User;
1920 import com.kob.backend.service.impl.utils.UserDetailsImpl;
1921 import com.kob.backend.service.oj.topic.UdataTopicService;
1922 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
1923 import org.springframework.security.core.context.SecurityContextHolder;
1924 import org.springframework.stereotype.Service;
1925
1926 import java.util.HashMap;
1927 import java.util.Map;
1928
1929 @Service
1930 public class UdataTopicServiceImpl implements UdataTopicService {
1931     private final TopicMapper topicMapper;
1932
1933     public UdataTopicServiceImpl(TopicMapper topicMapper) {
1934         this.topicMapper = topicMapper;
1935     }
1936
1937     @Override
1938     public Map<String, String> updateTopic(Map<String, String> data) {
1939         UsernamePasswordAuthenticationToken authenticationToken =
1940             (UsernamePasswordAuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
1941         UserDetailsImpl loginUser = (UserDetailsImpl) authenticationToken.getPrincipal();
1942         User user = loginUser.getUser();
1943
1944         HashMap<String,String> map=new HashMap<>();
1945
1946
1947         if(!user.getRoot().equals("true")){
1948             map.put("error_message", "你没有权限进行此操作");
1949             return map;
1950         }

```

第 40 页 | 行 1951-2000 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/oj/topic/UpdateTopicServiceImpl.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/user/account/InfoServiceImpl.java

```

1951         if(data.get("test_point")==null || data.get("test_point").equals("0")){
1952             map.put("error_message", "测试点数量不能为 0");
1953             return map;
1954         }
1955         if(data.get("title")==null || data.get("title").isEmpty()){
1956             map.put("error_message", "标题不能为空");
1957             return map;
1958         }
1959         if(data.get("title").length()>100){
1960             map.put("error_message", "标题的长度不能超过 100");
1961             return map;
1962         }
1963         if(data.get("description")==null || data.get("description").isEmpty()){
1964             map.put("error_message", "题目描述不能为空");
1965             return map;
1966         }
1967         if(data.get("time_limit") ==null || data.get("time_limit").equals("0")){
1968             map.put("error_message", "时间限制不能为 0");
1969             return map;
1970         }
1971         if(data.get("mem_limit") ==null || data.get("mem_limit").equals("0")){
1972             map.put("error_message", "内存限制不能为 0");
1973             return map;
1974         }
1975
1976         Topic new_topic=new Topic(
1977             Integer.parseInt(data.get("topic_id")),
1978             Integer.parseInt(data.get("test_point")),
1979             data.get("title"),
1980             data.get("description"),
1981             data.get("star"),
1982             Integer.parseInt(data.get("time_limit")),
1983             Integer.parseInt(data.get("mem_limit")),
1984             data.get("input_format"),
1985             data.get("output_format"),
1986             data.get("sample_input"),
1987             data.get("sample_output"),
1988             data.get("hint")
1989         );
1990         topicMapper.updateById(new_topic);
1991
1992         map.put("error_message", "success");
1993         return map;
1994     }
1995 }
1996 package com.kob.backend.service.impl.user.account;
1997
1998 import com.kob.backend.pojo.User;
1999 import com.kob.backend.service.impl.utils.UserDetailsImpl;
2000 import com.kob.backend.service.user.account.InfoService;

```


第 41 页 | 行 2001-2050 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/user/account/InfoServiceImp.java ...
 backendcloud/backend/src/main/java/com/kob/backend/service/impl/user/account/LoginServiceImp.java

```

2001 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
2002 import org.springframework.security.core.context.SecurityContextHolder;
2003 import org.springframework.stereotype.Service;
2004
2005 import java.util.Collections;
2006 import java.util.HashMap;
2007 import java.util.Map;
2008
2009 @Service
2010 public class InfoServiceImp implements InfoService {
2011
2012     @Override
2013     public Map<String, String> getinfo() {
2014         UsernamePasswordAuthenticationToken authentication =
2015             (UsernamePasswordAuthenticationToken) SecurityContextHolder.getContext().getAuthentication();
2016         UserDetailsImpl userDetails = (UserDetailsImpl) authentication.getPrincipal();
2017         User user = userDetails.getUser();
2018
2019         Map<String, String> map = new HashMap<>();
2020         if(user.getBanned().equals("true")){
2021             map.put("error_message", "该账号已被封禁");
2022             return map;
2023         }
2024         map.put("error_message", "success");
2025         map.put("id", user.getId().toString());
2026         map.put("username", user.getUsername());
2027         map.put("photo", user.getPhoto());
2028         map.put("root", user.getRoot());
2029         return map;
2030     }
2031 }
2032 package com.kob.backend.service.impl.user.account;
2033
2034 import com.kob.backend.pojo.User;
2035 import com.kob.backend.service.impl.utils.UserDetailsImpl;
2036 import com.kob.backend.service.user.account.LoginService;
2037 import com.kob.backend.utils.JwtUtil;
2038 import io.jsonwebtoken.Jwts;
2039 import org.springframework.beans.factory.annotation.Autowired;
2040 import org.springframework.security.authentication.AuthenticationManager;
2041 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
2042 import org.springframework.security.core.Authentication;
2043 import org.springframework.stereotype.Service;
2044
2045 import javax.swing.*;
2046 import java.util.Collections;
2047 import java.util.HashMap;
2048 import java.util.Map;
2049
2050 @Service

```

第 42 页 | 行 2051-2100 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/user/account/LoginServiceImpl.java ...
backendcloud/backend/src/main/java/com/kob/backend/service/impl/utlis/UserDetailsImpl.java

```
2051 public class LoginServiceImpl implements LoginService {
2052
2053     @Autowired
2054     private AuthenticationManager authenticationManager;
2055
2056     @Override
2057     public Map<String, String> getToken(String username, String password) {
2058         UsernamePasswordAuthenticationToken authenticationToken =
2059             new UsernamePasswordAuthenticationToken(username, password);
2060
2061         Authentication authentication = authenticationManager.authenticate(authenticationToken);
2062
2063         UserDetailsImpl loginUser = (UserDetailsImpl) authentication.getPrincipal();
2064         User user = loginUser.getUser();
2065         String jwt = JwtUtil.createJWT(user.getId().toString());
2066
2067         Map<String, String> map = new HashMap<>();
2068         if(user.getBanned().equals("true")){
2069             map.put("error_message", "该账号已被封禁");
2070             return map;
2071         }
2072         if(jwt == null){
2073             map.put("error_message", "账号或密码错误");
2074             return map;
2075         }
2076         map.put("error_message", "success");
2077         map.put("token", jwt);
2078         System.out.println(jwt);
2079
2080         return map;
2081     }
2082 }
2083 package com.kob.backend.service.impl.utlis;
2084
2085 import com.kob.backend.pojo.User;
2086 import lombok.AllArgsConstructor;
2087 import lombok.Data;
2088 import lombok.NoArgsConstructor;
2089 import org.springframework.security.core.GrantedAuthority;
2090 import org.springframework.security.core.userdetails.UserDetails;
2091
2092 import java.util.Collection;
2093 import java.util.Collections;
2094
2095 @Data
2096 @AllArgsConstructor
2097 @NoArgsConstructor
2098 public class UserDetailsImpl implements UserDetails {
2099
2100     private User user;
```

第 43 页 | 行 2101-2150 | backendcloud/backend/src/main/java/com/kob/backend/service/impl/utis/UserDetailsImpl.java ...
backendcloud/backend/src/main/java/com/kob/backend/service/manage/GitRecordListService.java

```
2101
2102
2103     @Override
2104     public Collection<? extends GrantedAuthority> getAuthorities() {
2105         return Collections.emptyList();
2106     }
2107
2108     @Override
2109     public String getPassword() {
2110         return user.getPassword();
2111     }
2112
2113     @Override
2114     public String getUsername() {
2115         return user.getUsername();
2116     }
2117
2118     @Override
2119     public boolean isAccountNonExpired() {
2120         return true;
2121     }
2122
2123     @Override
2124     public boolean isAccountNonLocked() {
2125         return true;
2126     }
2127
2128     @Override
2129     public boolean isCredentialsNonExpired() {
2130         return true;
2131     }
2132
2133     @Override
2134     public boolean isEnabled() {
2135         return true;
2136     }
2137 }
2138 package com.kob.backend.service.manage;
2139
2140 import com.kob.backend.pojo.Evaluate;
2141
2142 import java.util.List;
2143
2144 public interface GetEvaluateService {
2145     List<Evaluate> getEvaluateList(Integer topic_id);
2146 }
2147 package com.kob.backend.service.manage;
2148
2149 import com.kob.backend.pojo.LoginRecord;
2150
```

第 44 页 | 行 2151-2200 | backendcloud/backend/src/main/java/com/kob/backend/service/manage/GitRecordListService.java ...
backendcloud/backend/src/main/java/com/kob/backend/service/oj/evaluate/AddEvaluateService.java

```
2151 import java.util.List;
2152
2153 public interface GitRecordListService {
2154     List<LoginRecord> getGitRecordList();
2155 }
2156 package com.kob.backend.service.manage;
2157
2158 import com.kob.backend.pojo.User;
2159
2160 import java.util.HashMap;
2161 import java.util.List;
2162
2163 public interface GitUserListService {
2164     List<HashMap<String, String>> getuserlist();
2165 }
2166 package com.kob.backend.service.manage;
2167
2168 import java.util.Map;
2169
2170 public interface UpdataBannedService {
2171     Map<String, String> updataBanned(String username, Integer status);
2172 }
2173 package com.kob.backend.service.manage;
2174
2175 import java.util.List;
2176 import java.util.Map;
2177
2178 public interface UpdataEvaluateService {
2179     Map<String, String> updataevaluate(List<Map<String,String>> data);
2180 }
2181 package com.kob.backend.service.oj.answer;
2182
2183 import com.kob.backend.pojo.Answer;
2184
2185 import java.util.Map;
2186
2187 public interface AddAnswerService {
2188     Map<String,String> addAnswer(Map<String,String> data);
2189 }
2190 package com.kob.backend.service.oj.answer;
2191
2192 import java.util.Map;
2193
2194 public interface GetAnswerService {
2195     Map<String,String> getAnswer(Map<String,String> data);
2196 }
2197 package com.kob.backend.service.oj.evaluate;
2198
2199 import java.util.Map;
2200
```

第 45 页 | 行 2201-2250 | backendcloud/backend/src/main/java/com/kob/backend/service/oj/evaluate/AddEvaluateService.java ...
backendcloud/backend/src/main/java/com/kob/backend/service/oj/topic/GetTopicService.java

```
2201 public interface AddEvaluateService {
2202     String AddEvaluate(Map<String,String> data);
2203 }
2204 package com.kob.backend.service.oj.evaluate;
2205
2206 import com.alibaba.fastjson2.JSONObject;
2207
2208 public interface EndEvaluateService {
2209     String EndEvaluate(JSONObject jsonObject);
2210 }
2211 package com.kob.backend.service.oj.record;
2212
2213 import java.util.Map;
2214
2215 public interface AddRecordService {
2216     Map<String,String> add(Map<String,String> data);
2217 }
2218 package com.kob.backend.service.oj.record;
2219
2220 import com.kob.backend.pojo.RecordOfQuestion;
2221
2222 import java.util.List;
2223
2224 public interface GetListRecordService {
2225     List<RecordOfQuestion> getList();
2226 }
2227 package com.kob.backend.service.oj.topic;
2228
2229 import java.util.Map;
2230
2231 public interface AddTopicService {
2232     Map<String,String> add(Map<String,String> data);
2233 }
2234 package com.kob.backend.service.oj.topic;
2235
2236 import com.kob.backend.model.PageResponse;
2237 import com.kob.backend.pojo.Topic;
2238
2239 import java.util.List;
2240
2241 public interface GetListTopicService {
2242     List<Topic> getList();
2243
2244     PageResponse<Topic> getList(long page, String keyword);
2245 }
2246 package com.kob.backend.service.oj.topic;
2247
2248 import com.fasterxml.jackson.core.JsonProcessingException;
2249
2250 import java.util.Map;
```

第 46 页 | 行 2251-2300 | backendcloud/backend/src/main/java/com/kob/backend/service/oj/topic/GetTopicService.java ...
backendcloud/backend/src/main/java/com/kob/backend/utils/BloomInitializer.java

```
2251
2252 public interface GetTopicService {
2253     Map<String,String> getTopic(Integer topic_id) throws JsonProcessingException;
2254 }
2255 package com.kob.backend.service.oj.topic;
2256
2257 import java.util.Map;
2258
2259 public interface RemoveTopicService {
2260     Map<String,String> removeTopic(Map<String,String> data);
2261 }
2262 package com.kob.backend.service.oj.topic;
2263
2264 import java.util.Map;
2265
2266 public interface UpdataTopicService {
2267     Map<String,String> updateTopic(Map<String,String> map);
2268 }
2269 package com.kob.backend.service.user.account;
2270
2271 import org.springframework.web.multipart.MultipartFile;
2272
2273 import java.util.Map;
2274
2275 public interface AvatarService {
2276     Map<String, String> uploadAvatar(MultipartFile file);
2277 }
2278 package com.kob.backend.service.user.account;
2279
2280 import java.util.Map;
2281
2282 public interface InfoService {
2283     Map<String,String> getinfo();
2284 }
2285 package com.kob.backend.service.user.account;
2286
2287 import java.util.Map;
2288
2289 public interface LoginService {
2290     Map<String, String> getToken(String username, String password);
2291 }
2292 package com.kob.backend.service.user.account;
2293
2294 import java.util.Map;
2295
2296 public interface RegisterService {
2297     Map<String, String> register(String username, String password, String confirmedPassword);
2298 }
2299 package com.kob.backend.utils;
2300
```

```
2301 import com.google.common.hash.BloomFilter;
2302 import com.google.common.hash.Funnels;
2303 import com.kob.backend.mapper.BlogMapper;
2304 import com.kob.backend.mapper.TopicMapper;
2305 import com.kob.backend.mapper.UserMapper;
2306 import com.kob.backend.pojo.Blog;
2307 import com.kob.backend.pojo.Topic;
2308 import lombok.Getter;
2309 import org.springframework.beans.factory.annotation.Autowired;
2310 import org.springframework.boot.ApplicationArguments;
2311 import org.springframework.boot.ApplicationRunner;
2312 import org.springframework.stereotype.Component;
2313
2314 import java.nio.charset.StandardCharsets;
2315 import java.util.List;
2316
2317 @Component
2318 public class BloomInitializer implements ApplicationRunner {
2319
2320     @Getter
2321     private BloomFilter<Integer> bloomFilter_Blog =BloomFilter.create(
2322         Funnels.integerFunnel(),
2323         100000,
2324         0.01
2325     );
2326
2327
2328     @Getter
2329     private BloomFilter<Integer> bloomFilter_Topic = BloomFilter.create(
2330         Funnels.integerFunnel(),
2331         100000,
2332         0.01
2333     );
2334
2335     @Autowired
2336     private BlogMapper blogMapper;
2337
2338     @Autowired
2339     private TopicMapper topicMapper;
2340
2341     @Override
2342     public void run(ApplicationArguments args) throws Exception {
2343         List<Blog> list_blog = blogMapper.selectList(null);
2344         List<Topic> list_topic = topicMapper.selectList(null);
2345
2346         for(Blog blog : list_blog){
2347             bloomFilter_Blog.put(blog.getId());
2348         }
2349         for(Topic topic : list_topic){
2350             bloomFilter_Topic.put(topic.getId());
2351         }
2352     }
2353 }
```

第 48 页 | 行 2351-2400 | backendcloud/backend/src/main/java/com/kob/backend/utlis/BloomInitializer.java ...
backendcloud/backend/src/main/java/com/kob/backend/utlis/JwtUtil.java

```
2351     }
2352
2353 }
2354 package com.kob.backend.utlis;
2355
2356 import io.jsonwebtoken.Claims;
2357 import io.jsonwebtoken.JwtBuilder;
2358 import io.jsonwebtoken.Jwts;
2359 import io.jsonwebtoken.SignatureAlgorithm;
2360 import org.springframework.stereotype.Component;
2361
2362 import javax.crypto.SecretKey;
2363 import javax.crypto.spec.SecretKeySpec;
2364 import java.util.Base64;
2365 import java.util.Date;
2366 import java.util.UUID;
2367
2368 @Component
2369 public class JwtUtil {
2370     public static final long JWT_TTL = 60 * 60 * 1000L * 24 * 14; // 有效期 14 天
2371     public static final String JWT_KEY = "SDFGjhdsfalshdfHFdsjksdfs121232131afasdfac";
2372
2373     public static String getUUID() {
2374         return UUID.randomUUID().toString().replaceAll("-", "");
2375     }
2376
2377     public static String createJWT(String subject) {
2378         JwtBuilder builder = getJwtBuilder(subject, null, getUUID());
2379         return builder.compact();
2380     }
2381
2382     private static JwtBuilder getJwtBuilder(String subject, Long ttlMillis, String uuid) {
2383         SignatureAlgorithm signatureAlgorithm = SignatureAlgorithm.HS256;
2384         SecretKey secretKey = generalKey();
2385         long nowMillis = System.currentTimeMillis();
2386         Date now = new Date(nowMillis);
2387         if (ttlMillis == null) {
2388             ttlMillis = JwtUtil.JWT_TTL;
2389         }
2390
2391         long expMillis = nowMillis + ttlMillis;
2392         Date expDate = new Date(expMillis);
2393         return Jwts.builder()
2394             .setId(uuid)
2395             .setSubject(subject)
2396             .setIssuer("sg")
2397             .setIssuedAt(now)
2398             .signWith(signatureAlgorithm, secretKey)
2399             .setExpiration(expDate);
2400     }
```


第 49 页 | 行 2401-2450 | backendcloud/backend/src/main/java/com/kob/backend/utlis/JwtUtil.java ...
 backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/EvaluateSystemApplication.java

```

2401
2402     public static SecretKey generalKey() {
2403         byte[] encodeKey = Base64.getDecoder().decode(JwtUtil.JWT_KEY);
2404         return new SecretKeySpec(encodeKey, 0, encodeKey.length, "HmacSHA256");
2405     }
2406
2407     public static Claims parseJWT(String jwt) throws Exception {
2408         SecretKey secretKey = generalKey();
2409         return Jwts.parserBuilder()
2410             .setSigningKey(secretKey)
2411             .build()
2412             .parseClaimsJws(jwt)
2413             .getBody();
2414     }
2415 }
2416 package com.kob.backend.utlis;
2417
2418 public final class PaginationUtils {
2419     private PaginationUtils() {
2420     }
2421
2422     public static long parsePage(String page) {
2423         if (page == null) {
2424             return 1L;
2425         }
2426         try {
2427             long parsed = Long.parseLong(page.trim());
2428             return parsed > 0L ? parsed : 1L;
2429         } catch (NumberFormatException exception) {
2430             return 1L;
2431         }
2432     }
2433
2434     public static String normalizeKeyword(String keyword) {
2435         return keyword == null ? "" : keyword.trim();
2436     }
2437 }
2438 package com.evaluatesystem;
2439
2440
2441 import com.evaluatesystem.service.utlis.EvaluatePool;
2442 import org.springframework.beans.factory.annotation.Autowired;
2443 import org.springframework.boot.SpringApplication;
2444 import org.springframework.boot.autoconfigure.SpringBootApplication;
2445
2446 @SpringBootApplication
2447 public class EvaluateSystemApplication {
2448     public static void main(String[] args) {
2449         SpringApplication.run(EvaluateSystemApplication.class, args);
2450     }

```

第 50 页 | 行 2451-2500 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/EvaluateSystemApplication.java ...
 backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/config/RabbitmqConfig.java

```

2451 }
2452 package com.evaluatesystem.config;
2453
2454 import org.springframework.amqp.support.converter.Jackson2JsonMessageConverter;
2455 import org.springframework.amqp.support.converter.MessageConverter;
2456 import org.springframework.context.annotation.Bean;
2457 import org.springframework.context.annotation.Configuration;
2458
2459 @Configuration
2460 public class RabbitMqConfiguration {
2461
2462     @Bean
2463     public MessageConverter messageConverter(){
2464         return new Jackson2JsonMessageConverter();
2465     }
2466 }
2467
2468 package com.evaluatesystem.config;
2469
2470 import org.slf4j.Logger;
2471 import org.slf4j.LoggerFactory;
2472 import org.springframework.amqp.core.*;
2473 import org.springframework.amqp.rabbit.config.SimpleRabbitListenerContainerFactory;
2474 import org.springframework.amqp.rabbit.connection.CachingConnectionFactory;
2475 import org.springframework.amqp.rabbit.connection.CorrelationData;
2476 import org.springframework.amqp.rabbit.core.RabbitTemplate;
2477 import org.springframework.amqp.support.converter.Jackson2JsonMessageConverter;
2478 import org.springframework.beans.factory.annotation.Autowired;
2479 import org.springframework.boot.autoconfigure.amqp.SimpleRabbitListenerContainerFactoryConfigurer;
2480 import org.springframework.context.annotation.Bean;
2481 import org.springframework.context.annotation.Configuration;
2482
2483 @Configuration
2484 public class RabbitmqConfig {
2485
2486     private static final Logger log = LoggerFactory.getLogger(RabbitmqConfig.class);
2487
2488     @Autowired
2489     private CachingConnectionFactory connectionFactory;
2490
2491     @Autowired
2492     private SimpleRabbitListenerContainerFactoryConfigurer factoryConfigurer;
2493
2494     // ===== 队列 / 交换机 / 绑定 =====
2495     // 两个服务声明完全一致，谁先启动谁创建，后启动的复用
2496
2497     @Bean
2498     public Queue evaluateTaskQueue() {
2499         return QueueBuilder.durable("evaluate.task.queue").build();
2500     }

```

第 51 页 | 行 2501-2550 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/config/RabbitmqConfig.java

```

2501
2502     @Bean
2503     public DirectExchange evaluateTaskExchange() {
2504         return new DirectExchange("evaluate.task.exchange", true, false);
2505     }
2506
2507     @Bean
2508     public Binding evaluateTaskBinding() {
2509         return BindingBuilder.bind(evaluateTaskQueue())
2510             .to(evaluateTaskExchange()).with("evaluate.task");
2511     }
2512
2513     @Bean
2514     public Queue evaluateResultQueue() {
2515         return QueueBuilder.durable("evaluate.result.queue").build();
2516     }
2517
2518     @Bean
2519     public DirectExchange evaluateResultExchange() {
2520         return new DirectExchange("evaluate.result.exchange", true, false);
2521     }
2522
2523     @Bean
2524     public Binding evaluateResultBinding() {
2525         return BindingBuilder.bind(evaluateResultQueue())
2526             .to(evaluateResultExchange()).with("evaluate.result");
2527     }
2528
2529     // ===== 消费者工厂 =====
2530
2531     /**
2532      * 单消费者 + 手动 ACK
2533      * 适用场景: Evaluatesystem 消费判题任务 (Docker 资源有限, 一次只判一道)
2534      */
2535     @Bean
2536     public SimpleRabbitListenerContainerFactory singleListenerContainer() {
2537         SimpleRabbitListenerContainerFactory factory = new SimpleRabbitListenerContainerFactory();
2538         factory.setConnectionFactory(connectionFactory);
2539         factory.setMessageConverter(new Jackson2JsonMessageConverter());
2540         factory.setConcurrentConsumers(1);
2541         factory.setMaxConcurrentConsumers(1);
2542         factory.setPrefetchCount(1);
2543         factory.setAcknowledgeMode(AcknowledgeMode.MANUAL);
2544         return factory;
2545     }
2546
2547     /**
2548      * 多消费者 + 自动 ACK (并发/预取数可调)
2549      * 适用场景: Backend 消费判题结果 (轻量写库 + WebSocket 推送, 支持并发)
2550      */

```

第 52 页 | 行 2551-2600 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/config/RabbitmqConfig.java ...
 backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/Consumer.java

```

2551     @Bean
2552     public SimpleRabbitListenerContainerFactory multiListenerContainer() {
2553         SimpleRabbitListenerContainerFactory factory = new SimpleRabbitListenerContainerFactory();
2554         factoryConfigurer.configure(factory, connectionFactory); // 继承 Spring Boot 默认值
2555         factory.setMessageConverter(new Jackson2JsonMessageConverter());
2556         factory.setAcknowledgeMode(AcknowledgeMode.AUTO);
2557         return factory;
2558     }
2559
2560     // ===== RabbitTemplate (生产者端) =====
2561
2562     @Bean
2563     public RabbitTemplate rabbitTemplate() {
2564         RabbitTemplate template = new RabbitTemplate(connectionFactory);
2565         template.setMandatory(true);
2566         template.setMessageConverter(new Jackson2JsonMessageConverter());
2567
2568         // 消息到达 Broker 的回调
2569         template.setConfirmCallback((CorrelationData cd, boolean ack, String cause) -> {
2570             if (!ack) {
2571                 log.error("消息发送失败: correlationData={}, cause={}", cd, cause);
2572             }
2573         });
2574
2575         // 消息无法路由的回调 (Mandatory=true 才有)
2576         template.setReturnsCallback(returned -> {
2577             log.error("消息路由失败: exchange={}, route={}, replyCode={}, replyText={}",
2578                 returned.getExchange(), returned.getRoutingKey(),
2579                 returned.getReplyCode(), returned.getReplyText());
2580         });
2581
2582         return template;
2583     }
2584 }
2585 package com.evaluatesystem.service;
2586
2587 import com.alibaba.fastjson2.JSONObject;
2588
2589 import java.util.Map;
2590
2591 public interface EvaluateService {
2592     String Evaluate(JSONObject jsonObject);
2593 }
2594 package com.evaluatesystem.service.utlis;
2595
2596
2597 import cn.hutool.core.date.StopWatch;
2598 import cn.hutool.core.io.FileUtil;
2599 import cn.hutool.core.util.ArrayUtil;
2600 import com.alibaba.fastjson2.JSONObject;

```

第 53 页 | 行 2601-2650 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/Consumer.java

```
2601 import com.evaluatesystem.config.RabbitMqConfiguration;
2602 import com.github.dockerjava.api.DockerClient;
2603 import com.github.dockerjava.api.async.ResultCallback;
2604 import com.github.dockerjava.api.command.*;
2605 import com.github.dockerjava.api.model.*;
2606 import com.github.dockerjava.core.DockerClientBuilder;
2607 import org.springframework.amqp.rabbit.core.RabbitTemplate;
2608 import org.springframework.beans.factory.annotation.Autowired;
2609 import org.springframework.stereotype.Component;
2610 import org.springframework.web.client.RestTemplate;
2611
2612 import javax.swing.*;
2613 import java.io.Closeable;
2614 import java.io.File;
2615 import java.io.IOException;
2616 import java.nio.charset.StandardCharsets;
2617 import java.util.ArrayList;
2618 import java.util.List;
2619 import java.util.UUID;
2620 import java.util.concurrent.TimeUnit;
2621
2622 @Component
2623 public class Consumer {
2624
2625     public static RabbitTemplate rabbitTemplate;
2626
2627     @Autowired
2628     public void setRabbitTemplate(RabbitTemplate rabbitTemplate) {
2629         Consumer.rabbitTemplate = rabbitTemplate;
2630     }
2631
2632     private final String EvaluatePassUrl = "http://127.0.0.1:3000/oj/evaluate/end/";
2633
2634     private void EvaluatePass(JSONObject jsonObject) {
2635         rabbitTemplate.convertAndSend("evaluate.result.exchange", "evaluate.result", jsonObject);
2636         System.out.println(jsonObject.get("user_id") + "EvaluatePass Success");
2637     }
2638
2639     private String EvaluateFile(String code) {
2640
2641         String userDir = System.getProperty("user.dir");
2642
2643         String globalCodePathName = userDir + File.separator + "globalCode";
2644
2645         if(!FileUtil.exist(globalCodePathName)){
2646             FileUtil.mkdir(globalCodePathName);
2647         }
2648
2649         String userCodeParenPath = globalCodePathName + File.separator + UUID.randomUUID();
2650     }
```

[第 54 页 | 行 2651-2700](#) | [backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/Consumer.java](#)

```
2651     String userCodePath = userCodeParenPath + File.separator + "Main.java";
2652
2653     File userCodeFile = FileUtil.writeString(code, userCodePath, StandardCharsets.UTF_8);
2654     return userCodeParenPath;
2655 }
2656
2657 public EvaluateMessage compileFile(File userCodeFile) {
2658     String compileCmp = String.format("javac -encoding utf-8 -source 8 -target 8 %s", userCodeFile.getAbsolutePath());
2659
2660     try {
2661         Process compileProcess = Runtime.getRuntime().exec(compileCmp);
2662
2663         EvaluateMessage evaluateMessage = ProcessUtils.runProcessAndGetMessage(compileProcess, "编译");
2664
2665         if(evaluateMessage.getExitValue() != 0){
2666             evaluateMessage.setErrorMessage("编译错误");
2667         }
2668
2669         return evaluateMessage;
2670     } catch (IOException e) {
2671         throw new RuntimeException(e);
2672     }
2673 }
2674
2675 private String addDocker(String language, String userCodeParenPath) {
2676     DockerClient dockerClient = DockerClientBuilder.getInstance().build();
2677     String image = "";
2678     boolean imageExists = false;
2679     if(language.equals("java")){ // 根据语言不同, 选择不一样的 docker 镜像下载
2680         image = "eclipse-temurin:8-jdk";
2681     }
2682     try {
2683         dockerClient.inspectImageCmd(image).exec();
2684         imageExists = true;
2685     }
2686     catch (Exception e) {
2687         System.out.println("镜像不存在, 开始下载 docker 镜像");
2688     }
2689     if(!imageExists){
2690         PullImageCmd pullImageCmd = dockerClient.pullImageCmd(image);
2691         PullImageResultCallback pullImageResultCallback = new PullImageResultCallback() {
2692             @Override
2693             public void onNext(PullResponseItem item){
2694                 System.out.println("下载镜像:" + item.getStatus());
2695                 super.onNext(item);
2696             }
2697         };
2698     };
2699
2700     try {
```

第 55 页 | 行 2701-2750 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/Consumer.java

```

2701     pullImageCmd.exec(pullImageResultCallback).awaitCompletion();
2702     } catch (InterruptedException e) {
2703         System.out.println("拉取镜像错误");
2704         throw new RuntimeException(e);
2705     }
2706 }
2707 CreateContainerCmd containerCmd = dockerClient.createContainerCmd(image);
2708
2709 HostConfig hostConfig = new HostConfig();
2710 hostConfig.withMemory(100 * 1000 * 1000L);
2711 hostConfig.withMemorySwap(0L);
2712 hostConfig.withCpuCount(1L);
2713 hostConfig.setBinds(new Bind(userCodeParenPath, new Volume("/app")));
2714
2715 CreateContainerResponse createContainerResponse = containerCmd
2716     .withHostConfig(hostConfig)
2717     .withNetworkDisabled(true)           // 禁用网络
2718     .withReadOnlyRootfs(true)           // 只读根文件系统
2719     .withAttachStdin(true)
2720     .withAttachStderr(true)
2721     .withAttachStdout(true)
2722     .withCmd("sh", "-c", "while true; do sleep 3600; done")
2723     .withTty(true)
2724     .exec();
2725 String container_id = createContainerResponse.getId();
2726 dockerClient.startContainerCmd(createContainerResponse.getId()).exec();
2727 return container_id;
2728 }
2729
2730 private List<EvaluateMessage> addEvaluate(List<String> inputList, String container_id){
2731     DockerClient dockerClient = DockerClientBuilder.getInstance().build();
2732     List<EvaluateMessage> evaluateList = new ArrayList<>();
2733     for(String input : inputList) {
2734         Stopwatch stopWatch = new Stopwatch();
2735         String escapedInput = input
2736             .replace("\\", "\\")
2737             .replace("'", "'")
2738             + "\\n";
2739
2740         String runCommand = "printf '" + escapedInput + "' | java -cp /app Main";
2741         ExecCreateCmdResponse execCreateCmdResponse = dockerClient.execCreateCmd(container_id)
2742             .withCmd("sh", "-c", runCommand)
2743             .withAttachStderr(true)
2744             .withAttachStdout(true)
2745             .withAttachStdin(true)
2746             .withAttachStdout(true)
2747             .exec();
2748         Long maxTime = 0L;
2749         final Long[] maxMemory = {0L};
2750         String exec_id = execCreateCmdResponse.getId();

```

```

2751
2752     DockerOutputCollector outputCollector = new DockerOutputCollector();
2753     StatsCmd statsCmd = dockerClient.statsCmd(container_id);
2754     ResultCallback<Statistics> resultCallback = statsCmd.exec(
2755         new ResultCallback<Statistics>() {
2756             @Override
2757             public void close() throws IOException {
2758             }
2759
2760             @Override
2761             public void onNext(Statistics statistics) {
2762                 maxMemory[0] = Math.max(maxMemory[0], statistics.getMemoryStats().getUsage());
2763             }
2764
2765             @Override
2766             public void onStart(Closeable closeable) {
2767             }
2768
2769             @Override
2770             public void onError(Throwable throwable) {
2771             }
2772
2773             @Override
2774             public void onComplete() {
2775             }
2776         });
2777
2778     try {
2779         stopWatch.start();
2780
2781         dockerClient.execStartCmd(exec_id)
2782             .exec(outputCollector)
2783             .awaitCompletion(2000, TimeUnit.MILLISECONDS);
2784
2785         stopWatch.stop();
2786
2787         maxTime = Math.max(maxTime, stopWatch.getTotalTimeMillis());
2788     } catch (InterruptedException e) {
2789         throw new RuntimeException(e);
2790     } finally {
2791         statsCmd.close();
2792     }
2793     EvaluateMessage evaluateMessage = new EvaluateMessage();
2794     evaluateMessage.setErrorMessage(outputCollector.getStderr());
2795     evaluateMessage.setMessage(outputCollector.getStdout());
2796     evaluateMessage.setTime(maxTime);
2797     evaluateMessage.setMemory(maxMemory[0]);
2798
2799     evaluateList.add(evaluateMessage);
2800 }

```


第 57 页 | 行 2801-2850 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/Consumer.java

```

2801     dockerClient.stopContainerCmd(container_id)
2802         .withTimeout(0)
2803         .exec();
2804     dockerClient.removeContainerCmd(container_id).exec();
2805     return evaluateList;
2806 }
2807
2808
2809
2810 public void startEvaluate(JSONObject jsonObject) {
2811     Integer user_id = jsonObject.getInteger("user_id");
2812     Integer evaluate_id = jsonObject.getInteger("evaluate_id");
2813     String code = jsonObject.getString("code");
2814     String language = jsonObject.getString("language");
2815     List<String> inputList = jsonObject.getJSONArray("inputList").toJavaList(String.class);
2816     List<String> outputList = jsonObject.getJSONArray("outputList").toJavaList(String.class);
2817     Integer quantity = inputList.size();
2818
2819     JSONObject message = new JSONObject();
2820     message.put("user_id", user_id);
2821     message.put("evaluate_id", evaluate_id);
2822
2823     String userCodeParenPath = EvaluateFile(code);//将用户代码打包成文件
2824
2825     File userCodeFile = new File(userCodeParenPath + File.separator + "Main.java");
2826     EvaluateMessage evaluateMessage = compileFile(userCodeFile);//将用户代码进行编译
2827     if(evaluateMessage.getErrorMessage().equals("编译错误")){
2828         message.put("state", "Compile Error");
2829         message.put("score", 0);
2830         EvaluatePass(message);
2831         return;
2832     }
2833
2834     String container_id = addDocker(language, userCodeParenPath);//检测用户判题的语言的 docker 镜像是否存在，如果不存在则下载
2835
2836     List<EvaluateMessage> evaluateMessagesList = addEvaluate(inputList, container_id);// 使用 docker 容器开始判题，并且将判题的结果返回
2837
2838     Integer score = 0;
2839     String state = "Wrong Answer";
2840     for(int i = 0; i < evaluateMessagesList.size(); i++){
2841         EvaluateMessage evaluateMessage1 = evaluateMessagesList.get(i);
2842         String output = outputList.get(i);
2843         String input = evaluateMessage1.getMessage();
2844         String error = evaluateMessage1.getErrorMessage();
2845         Long time = evaluateMessage1.getTime();
2846         Long memory = evaluateMessage1.getMemory();
2847         input = input.replace("\n", "");
2848
2849         if(error != null && !error.equals("")){
2850             message.put("state", "Runtime Error");

```

第 58 页 | 行 2851-2900 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/Consumer.java ...
 backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/DockerOutputCollector.java

```

2851         message.put("score", 0);
2852         EvaluatePass(message);
2853         return;
2854     }
2855     if(output.equals(input)) score += 1;
2856 }
2857 if(quantity != 0) {
2858     score = (score / quantity) * 100;
2859 }
2860 else {
2861     score = 0;
2862 }
2863 if (score == 100) {
2864     message.put("state", "Accepted");
2865     message.put("score", score);
2866 }
2867 else if (score < 100) {
2868     message.put("state", "Wrong Answer");
2869     message.put("score", score);
2870 }
2871 System.out.println(user_id + " " + evaluate_id + " " + state + " " + score);
2872 EvaluatePass(message);
2873 FileUtil.del(userCodeParenPath);
2874 }
2875 }
2876 package com.evaluatesystem.service.utlis;
2877
2878 import com.github.dockerjava.core.command.ExecStartResultCallback;
2879
2880 import java.io.ByteArrayOutputStream;
2881 import java.nio.charset.StandardCharsets;
2882
2883 final class DockerOutputCollector extends ExecStartResultCallback {
2884     private final ByteArrayOutputStream stdout;
2885     private final ByteArrayOutputStream stderr;
2886
2887     DockerOutputCollector() {
2888         this(new ByteArrayOutputStream(), new ByteArrayOutputStream());
2889     }
2890
2891     private DockerOutputCollector(ByteArrayOutputStream stdout, ByteArrayOutputStream stderr) {
2892         super(stdout, stderr);
2893         this.stdout = stdout;
2894         this.stderr = stderr;
2895     }
2896
2897     String getStdout() {
2898         return new String(stdout.toByteArray(), StandardCharsets.UTF_8);
2899     }
2900 }

```

第 59 页 | 行 2901-2950 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/DockerOutputCollector.java ...
 backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/EvaluatePool.java

```

2901     String getStderr() {
2902         return new String(stderr.toByteArray(), StandardCharsets.UTF_8);
2903     }
2904 }
2905 package com.evaluatesystem.service.utlis;
2906
2907 import lombok.AllArgsConstructor;
2908 import lombok.Data;
2909 import lombok.NoArgsConstructor;
2910
2911 @Data
2912 @AllArgsConstructor
2913 @NoArgsConstructor
2914 public class EvaluateMessage {
2915     private Integer exitValue;
2916     private String message;
2917     private String errorMessage;
2918     private Long time;
2919     private Long memory;
2920 }
2921 package com.evaluatesystem.service.utlis;
2922
2923 import com.alibaba.fastjson2.JSONObject;
2924 import org.springframework.amqp.rabbit.annotation.RabbitListener;
2925 import org.springframework.amqp.rabbit.core.RabbitTemplate;
2926 import org.springframework.beans.factory.annotation.Autowired;
2927 import org.springframework.boot.context.event.ApplicationReadyEvent;
2928 import org.springframework.context.event.EventListener;
2929 import org.springframework.stereotype.Component;
2930
2931 import java.util.Queue;
2932 import java.util.concurrent.BlockingQueue;
2933 import java.util.concurrent.ConcurrentLinkedQueue;
2934 import java.util.concurrent.LinkedBlockingQueue;
2935 import java.util.concurrent.locks.Condition;
2936 import java.util.concurrent.locks.Lock;
2937 import java.util.concurrent.locks.ReentrantLock;
2938
2939 @Component
2940 public class EvaluatePool {
2941     @Autowired
2942     private final Consumer consumer;
2943
2944     public EvaluatePool(Consumer consumer) {
2945         this.consumer = consumer;
2946     }
2947
2948     @RabbitListener(queues = "evaluate.task.queue", concurrency = "2")
2949     private void pool(JSONObject message) {
2950         consumer.startEvaluate(message);

```

第 60 页 | 行 2951-3000 | backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/EvaluatePool.java ...
backendcloud/evaluatesystem/src/main/java/com/evaluatesystem/service/utlis/ProcessUtils.java

```
2951     }
2952 }
2953 package com.evaluatesystem.service.utlis;
2954
2955 import java.io.BufferedReader;
2956 import java.io.IOException;
2957 import java.io.InputStream;
2958 import java.io.InputStreamReader;
2959 import java.nio.charset.StandardCharsets;
2960
2961 public class ProcessUtils {
2962
2963     public static EvaluateMessage runProcessAndGetMessage(Process process, String operationName) {
2964         long startTime = System.currentTimeMillis();
2965         StringBuilder message = new StringBuilder();
2966         StringBuilder errorMessage = new StringBuilder();
2967
2968         Thread outputThread = new Thread(() -> readStream(process.getInputStream(), message));
2969         Thread errorThread = new Thread(() -> readStream(process.getErrorStream(), errorMessage));
2970         outputThread.start();
2971         errorThread.start();
2972
2973         try {
2974             int exitValue = process.waitFor();
2975             outputThread.join();
2976             errorThread.join();
2977
2978             EvaluateMessage evaluateMessage = new EvaluateMessage();
2979             evaluateMessage.setExitValue(exitValue);
2980             evaluateMessage.setMessage(message.toString());
2981             evaluateMessage.setErrorMessage(errorMessage.toString());
2982             evaluateMessage.setTime(System.currentTimeMillis() - startTime);
2983             return evaluateMessage;
2984         } catch (InterruptedException e) {
2985             Thread.currentThread().interrupt();
2986             throw new RuntimeException(operationName + "进程被中断", e);
2987         }
2988     }
2989
2990     private static void readStream(InputStream inputStream, StringBuilder result) {
2991         try (BufferedReader reader = new BufferedReader(new InputStreamReader(inputStream, StandardCharsets.UTF_8))) {
2992             String line;
2993             while ((line = reader.readLine()) != null) {
2994                 result.append(line).append(System.lineSeparator());
2995             }
2996         } catch (IOException e) {
2997             throw new RuntimeException("读取进程输出失败", e);
2998         }
2999     }
3000 }
```